

Single-phase immersion cooling for data centers and HPC: Recent advances and system-level challenges

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ABSTRACT

Single-phase immersion cooling (SPIC) has emerged as a promising solution to address the escalating thermal demands of data centers and high-performance computing (HPC). This review consolidates recent advances in single-phase immersion cooling system, including discussions on dielectric fluids, cooling architectures, and intelligent control frameworks, with emphasis on experimental validation and system-level deployment. Progress in dielectric fluid is also addressed in details. The common dielectric fluid includes synthetic hydrocarbons and nanofluid-enhanced oils that were examined in association with thermal performance and compatibility with electronic components. At the control frontier, models subject to predictive control, digital twins, and reinforcement learning have enabled real-time optimization, predictive fault detection, and adaptive energy management. In parallel, sustainability metrics such as Carbon Usage Effectiveness (CUE), Water Usage Effectiveness (WUE), Renewable Energy Factor (REF), and Life Cycle Assessment (LCA) extend evaluation beyond traditional power usage effectiveness (PUE), incorporating water footprint, carbon impact, and waste heat recovery. Comparative assessments confirm SPIC's operational stability and scalability, but also highlight persistent challenges in coolant reliability, system retrofitting, and certification standards. By bridging the fluid/thermal science, intelligent control, and sustainability metrics, this review provides a comprehensive framework for advancing SPIC as a cornerstone of sustainable, scalable data center cooling.

1. Introduction

The accelerating demand for computing power, driven by artificial intelligence (AI), high-performance computing (HPC), cloud services, and crypto mining, has pushed conventional air-cooling technologies to their practical limits. Recent assessments highlight that modern AI accelerators, particularly H100-class GPUs, exhibit localized die-level heat fluxes of approximately 800–1000 kW/m² (80–100 W/cm²), while consumer GPUs such as the RTX 4090 exhibit die-scale heat fluxes of around 700 kW/m² (70 W/cm²) [1]. At the server level, maldistribution of coolant flow further compounds these thermal challenges, underscoring the need for optimized manifold and delivery architectures to maintain uniform cooling performance [2]. Meanwhile, advanced direct liquid and two-phase cooling solutions have demonstrated scalability toward ~2500 kW/m² (250 W/cm²), offering feasible pathways to manage next-generation accelerator loads in hyperscale deployments [3]. The urgency of these innovations is amplified by the exponential growth of transformer-based AI models and dense GPU clusters that underpin cloud-scale services [4].

In this context, single-phase immersion cooling (SPIC) has emerged as a promising alternative. By directly submerging electronic components in thermally conductive yet electrically insulating dielectric fluids, SPIC enables efficient convective heat transfer while eliminating the need for fans and heat sinks especially for multi heat sources (e.g. GPU + CPU). These features contribute to lower power usage effectiveness (PUE), reduced acoustic noise, and enhanced system reliability [5,6]. In addition, its mechanical simplicity and compatibility with diverse server architectures support large-scale deployment in modular rack-level and containerized data centers [7,8].

Recent studies confirmed these advantages while also pointing to remaining limitations. Zheng et al. [5] provided a comprehensive review of dielectric fluids, architectures, and material compatibility but gave limited attention to AI workloads and high-density environments. Building on this foundation, numerical analyses by Rongyang et al. [9] showed that thermal performance can be tuned through coolant selection and system geometry. Experimental investigations by Rokonzaman et al. [10] and Pambudi et al. [11] demonstrated superior heat transfer efficiency and energy savings over air cooling, while Ge et al.

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[12] and Wang et al. [13] compared forced and natural convection, highlighting trade-offs in thermal uniformity. At rack level, Kim et al. [14] reported up to 70% server temperature reduction under high-power-density conditions. Complementary work by Muneeshwaran et al. [8] and Dharanegowda and Gowda [15] assessed FC-40 and mineral oil under varying port and flow-rate configurations. Broader reviews further underscore the importance of improved chip-scale thermal strategies [16].

Beyond driving unprecedented demand, AI also plays an increasingly pivotal role in optimizing thermal management, with AI-based predictive control shown to improve energy efficiency in liquid-cooled and hybrid systems [17,18], underscoring the growing synergy between AI and thermal management. Nonetheless, barriers remain. These include coolant degradation, component compatibility, maintenance, and the absence of standardized performance benchmarks [19–22]. Long-term reliability also poses a critical challenge, with studies highlighting risks to PCB thermo-mechanical stability and package durability under immersion exposure [20,22]. More recent work emphasizes the need for systematic evaluation of server components and immersion fluids to ensure operational stability over extended lifetimes [21,23]. In parallel, sustainability considerations, hybrid integration with secondary loops, and facility-level readiness remain essential for large-scale adoption.

A visual comparison of cooling capacity across various thermal management strategies, ranging from air cooling to two-phase immersion cooling, against projected system-level heat flux demands for standard CPU servers, GPU training clusters, HPC nodes, and AI supercomputing workloads. Cooling capacity data are compiled from Muneeshwaran et al. [8], Ge et al. [12], Kim et al. [14], and projected workload demands from Jenkins and Robinson [1] and Rokonzaman et al. [10], clearly illustrate that while SPIC can accommodate many current high-density computing needs, only TPIC provides the thermal headroom projected for future AI supercomputing workloads. However, practical challenges associated with TPIC, such as the need for precise pressure control and the volatility of dielectric fluids, have limited its technological maturity and large-scale deployment. This contrast underscores the need for continued innovation across both immersion cooling paradigms. In this context, “AI workloads” refer to the broad class of GPU- and accelerator-based computing currently deployed in data centers, whereas “AI supercomputing workloads” denote future ultra-high-density training systems operating at extreme heat flux levels.

Given these drivers, opportunities, and challenges, this review deliberately focuses on SPIC, which occupies a critical position between thermally limited air and cold-plate cooling solutions and higher-capacity but less deployment-ready two-phase immersion systems. The novelty of this work lies in its system-level synthesis of SPIC performance and limitations under emerging AI and HPC workloads, integrating experimental results, numerical models, and deployment-scale considerations that are often treated separately in the literature. Section 2 introduces SPIC fundamentals and its relevance to AI/HPC; Section 3 reviews coolant materials and system design; Section 4 explores applications in data centers and HPC, including scalability and energy efficiency; Section 5 examines operational reliability and maintenance; Section 6 outlines major challenges in thermal uniformity, material compatibility, and system integration; and Section 7 discusses hybrid cooling, sustainable design, intelligent control, and standardization. Collectively, these sections provide a framework for advancing SPIC deployment in data centers, HPC, and AI-intensive environments.

2. Fundamentals of single-phase immersion cooling

Before discussing design strategies and system-level implementations, it is necessary to establish the physical operating principles and comparative context of SPIC. This section introduces SPIC's governing mechanisms (2.1) along with its benchmark against conventional and two-phase approaches (2.2), and highlights its unique relevance for emerging high-power computing workloads (2.3).

2.1. Fundamentals of SPIC

SPIC immerses electronic components in thermally conductive, electrically insulating dielectric fluids, removing heat through single-phase convection. Operating below the boiling point, SPIC avoids phase-change instabilities, enabling stable and predictable thermal performance. This is exemplified in a typical experimental setup (Fig. 2), which features a closed-loop configuration with a thermostatic bath, circulation pump, inline filter, flow regulator and integrated sensors for temperature and flow rate monitoring. The data server is fully immersed in the dielectric fluid, enabling direct convective heat transfer at the component–fluid interface.

Fluid stability is a critical reliability factor. Daccord et al. [26]

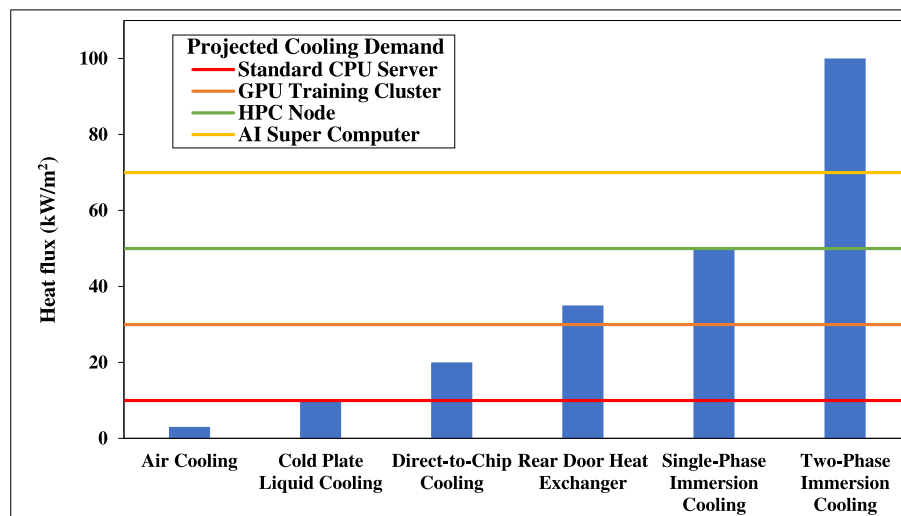


Fig. 1. Comparison of cooling capacities across major data center thermal management strategies, ranging from air cooling to two-phase immersion cooling, against projected system-level heat flux demands for standard CPU servers, GPU training clusters, HPC nodes, and AI supercomputing workloads. Cooling capacity data are compiled from Muneeshwaran et al. [8], Ge et al. [12], Kanbur et al. [24], and Kim et al. [14], while demand projections are sourced from Jenkins and Robinson [1] and Rokonzaman et al. [10]. Note: Projected heat fluxes reflect average system-level thermal loads distributed across the server or rack footprint. While SPIC approaches the upper limits of GPU and HPC workloads, two-phase immersion cooling provides substantially greater thermal headroom for projected AI supercomputing demands ($O(10^3 \text{ kW/m}^2)$).

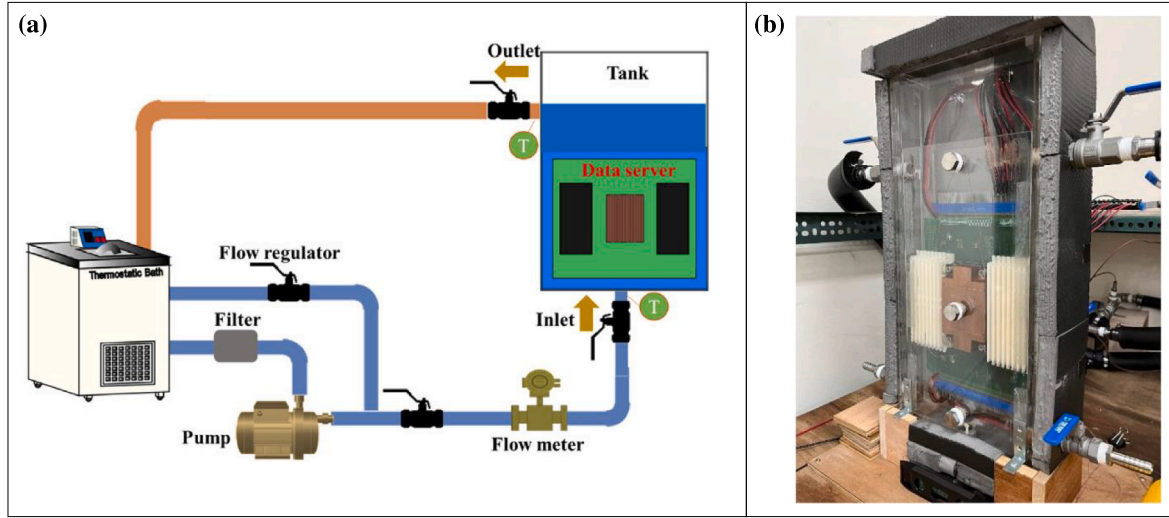


Fig. 2. (a) Schematic of the experimental setup for single-phase immersion cooling, showing the closed-loop flow system with a thermal bath, pump, filter, flow regulator, flow meter, and temperature sensors at the inlet and outlet of the server tank. (b) Photograph of the actual immersion cooling system with the transparent test rig housing the data server components. Figures adapted from Shrigondekar et al. [25].

showed that thermal aging reduces heat transfer coefficients and increases fluid acidity, underscoring the need for maintenance or replacement schedules. Unlike two-phase systems that facilitate nucleate boiling mechanism, hence effective vapor regulation, pressure control, purge of non-condensable gas, and condensation management are critical measure upon system design [27], the SPIC is inherited simpler than that of two-phase immersion cooling system.

SPIC also improves energy efficiency by reducing reliance on mechanical chillers and airflow systems. Chainer et al. [28] reported measurable gains in power usage effectiveness (PUE), while Chen and Li [29] emphasized its role in sustainable data center design. These advantages are corroborated by experimental findings. For instance, Huang et al. [30] demonstrated low inter-rack temperature gradients under variable workloads, while Shrigondekar et al. [25] observed

superior chip-level thermal stabilization compared to forced-air cooling.

Extending beyond data center applications, Li et al. [31] demonstrated that immersion cooling significantly improves thermal regulation in lithium-ion batteries by enhancing convective heat transfer through direct fluid contact with high surface-area cell walls. Complementing this, Liu et al. [32] systematically evaluated a range of dielectric fluids for single-phase immersion cooling, revealing that the thermal conductivity, specific heat, and viscosity of the working fluid critically influence cooling performance, particularly in maintaining cell uniformity and preventing thermal runaway.

Finally, SPIC enables energy reuse beyond primary heat dissipation. The iDataCool project [33] demonstrated that hot-water-based cooling, with outlet temperatures reaching up to 70 °C, can be effectively repurposed for building heating and adsorption chillers—contributing

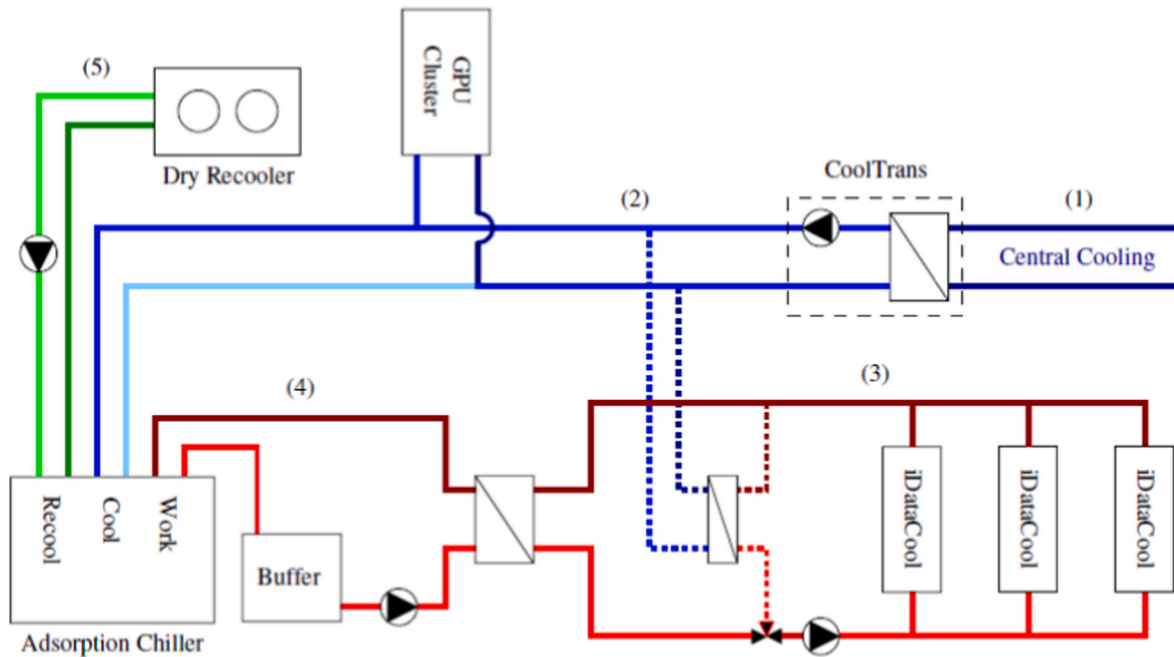


Fig. 3. Schematic of the iDataCool liquid cooling architecture, highlighting five interconnected circuits: (1) central facility cooling, (2) primary coolant supply, (3) rack-level immersion cooling of iDataCool servers, (4) secondary loop driving an adsorption chiller for energy reuse, and (5) dry recoler circuit. This configuration enables effective heat removal and supports hot-water reuse for building-level heating or cooling applications. Figure adapted from Meyer et al. [33].

to sustainable high-performance computing (HPC) operations. The closed-loop configuration, shown in Fig. 3, integrates multiple sub-systems: a central cooling circuit, rack-level immersion loops, a secondary driving circuit for heat recovery, and a dry cooler. This architecture underscores the broader system-level advantages of SPIC in achieving energy-efficient and thermally integrated data center infrastructures.

In essence, SPIC offers characteristics like thermal stability, energy efficiency, and versatility across application that make it a compelling choice for modern and future high-density electronic systems.

2.2. Comparison with other cooling methods

Where available, the following comparison is based on quantitative metrics reported in the literature, including operating temperature, temperature difference, power level, energy consumption, and heat flux capability.

Single-phase immersion cooling (SPIC) presents a distinct advantage over conventional air-cooling and more complex two-phase immersion cooling (TPIC) systems. Traditional air cooling, which dominates the majority of existing data centers, relies on forced convection using fans and air movers to dissipate heat through heat sinks. However, the inferior thermal conductivity of air results in a rather low heat transfer coefficient and limits the thermal capacity accordingly, which severely constrains its ability to manage high heat fluxes. As a result, air cooling suffers from significant thermal resistance, particularly at the interfaces between components and cooling hardware, and requires aggressive high airflow strategies which not only demand elevated energy consumption for fans but also raise severe noise concerns. The poor heat transfer performance of air-cooling along with comparatively high ambient temperatures, ultimately leading to poor energy efficiency of mechanical chillers and yields much higher operating costs.

In contrast, SPIC submerges server components directly into dielectric fluids, removing the need for intermediate heat exchangers and reducing parasitic power consumption. Multiple studies have shown that SPIC enables more uniform temperature profiles and supports higher power densities without overheating. For example, experimental data by Huang et al. [30] and Muneeshwaran et al. [8] confirmed that SPIC maintained GPU and CPU temperatures below 60 °C even under

peak loads, with minimal temperature fluctuation. Dharanegowda and Gowda [15] further reported that SPIC reduced overall server-level power consumption by up to 20% compared to air cooling due to the elimination of server fans.

TPIC enables high heat flux management due to latent heat transfer, potentially exceeding 1000 kW/m² (100 W/cm²) at chip-level hotspots [35], and system-level demonstrations [24] validate its feasibility for high-power-density racks. However, TPIC introduces significant operational challenges, including vapor handling, pressure management, fluid stability, and long-term reliability. Degradation of the working fluid can lead to fouling, corrosion, and performance deterioration over time [26]. Fig. 4 illustrates the complexity of maintaining TPIC systems, showing both the schematic and real-world setup used to simulate aging conditions. The experimental apparatus includes vapor containment, pressure sensors, check valves, desiccant air breathers, and thermal control for high-temperature operation (80 °C), highlighting the level of instrumentation and control required to ensure stable two-phase performance. These added layers of control contrast sharply with SPIC's robust operation, underscoring its lower maintenance burden and better suitability for real-world deployment.

In comparison, SPIC offers lower maintenance overhead while achieving acceptable thermal performance. Qiu et al. [34] reported that although the temperature difference of bulk-to-inlet (ΔT) in SPIC is moderately higher than in TPIC due to the lack of phase change, the absolute bulk fluid temperature remains well below thermal limits across various power outputs. As shown in Fig. 5, the T_{bulk} values fall below 55 °C for SPIC at an inlet temperature of 40 °C even at a higher power input of 2000 W. The ΔT remains under 12 °C, highlighting its capability to provide reliable cooling with a simpler system architecture. These data support the case for SPIC as a robust and scalable alternative where ease of deployment, maintenance, and energy reuse outweigh the added thermal advantages of TPIC. Shrigondekar et al. [25] also reported its modular scalability for rapid deployment. Reviews by Zheng et al. [5] and Chen and Li [29] similarly conclude that while air cooling remains cost-effective for low power densities and TPIC excels at extreme loads (e.g., AI training clusters), SPIC effectively supports medium-to-high density workloads with straightforward integration.

Finally, SPIC enhances sustainability by enabling high-grade heat reuse. Chainer et al. [28] and Qiu et al. [34] demonstrated that waste

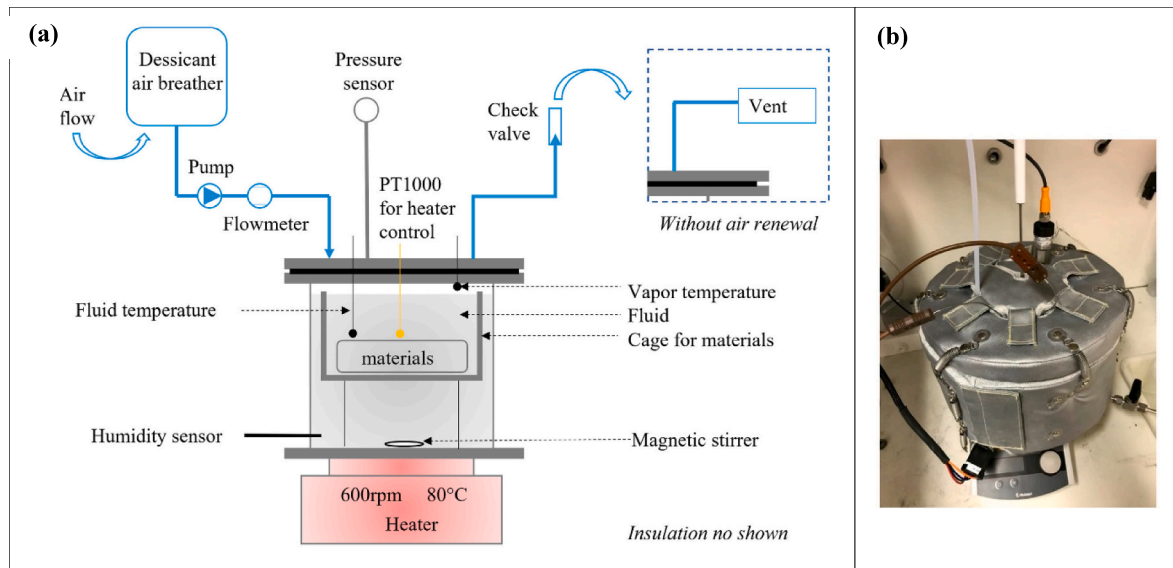


Fig. 4. Experimental setup for long-term TPIC fluid aging and degradation analysis. (a) Schematic of the TPIC aging loop developed by Daccord et al. [26], designed to simulate operational exposure of dielectric fluids under typical two-phase immersion conditions. (b) Photo of the actual setup, showing the thermally insulated reactor vessel and instrumentation required for stable, prolonged two-phase operation. This figure illustrates the operational overhead associated with TPIC, such as vapor control, pressure balancing, and degradation monitoring, reinforcing SPIC's system-level simplicity and lower maintenance burden in practical deployments. Figures adapted from Daccord et al. [26].

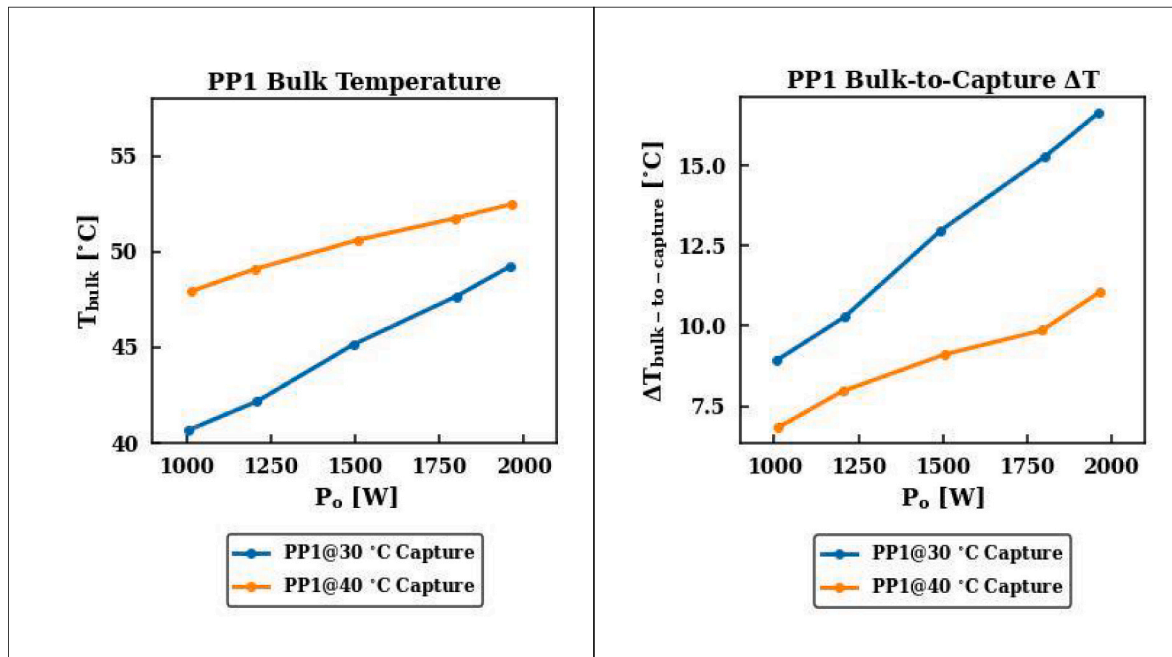


Fig. 5. Thermal comparison of SPIC and TPIC under equivalent power loads. (A) Bulk fluid temperature (T_{bulk}) and (B) bulk-to-capture temperature difference ($\Delta T_{bulk-to-capture}$) for single-phase (PP1@40 °C) and two-phase (PP1@30 °C) immersion cooling systems. Results show that while SPIC exhibits slightly higher ΔT due to the absence of latent heat transfer, it still maintains safe operating temperatures across tested power levels. Figures adapted from Qiu et al. [34], highlighting SPIC's acceptable thermal performance despite its simpler system architecture.

coolant can be repurposed for building heating, improving overall energy utilization in data center infrastructure.

Other advanced cooling approaches, such as single-phase and two-phase cold-plate cooling, are not discussed in detail here, as they represent intermediate solutions between direct-to-chip liquid cooling and full immersion systems and fall outside the primary scope of this immersion-focused comparison.

2.3. Thermal relevance to AI and HPC

Artificial intelligence (AI) and high-performance computing (HPC) workloads impose a distinct set of cooling requirements compared with conventional data center applications. These include (i) sustained operation under extremely high heat fluxes at both chip and rack levels, (ii) mitigation of strong spatial non-uniformity arising from GPUs, VRMs, and memory subsystems, (iii) rapid thermal response under highly transient workloads, and (iv) scalability to dense, modular deployment without excessive system complexity. The following discussion evaluates SPIC in the context of these requirements.

The growing thermal demands of HPC and AI workloads, especially driven by dense packaging, high power draw, and rapid transients from GPUs, TPUs, and memory components, necessitate advanced thermal management strategies. SPIC has emerged as a promising candidate to meet these needs by delivering spatially uniform heat removal, suppressing hotspots, and enabling simplified system design in modular and edge data centers.

One of the most pressing challenges in AI hardware is the spatial non-uniformity of power dissipation, with localized hotspots arising near high-activity zones such as GPUs and voltage regulation modules. Qiu et al. [34] demonstrated that single-phase immersion cooling (SPIC) mitigates these hotspots at the component level. Using infrared thermography of a power supply unit (PSU), they reported a 13.5 °C peak temperature reduction under SPIC relative to air cooling. More importantly, their deployment of a distributed sensor network across GPU, VRM, and PCB regions confirmed SPIC's ability to maintain fine-grained thermal uniformity which is a critical requirement for thermally dense,

multi-component AI systems.

These thermal improvements, however, cannot be fully leveraged without high spatial and temporal resolution diagnostics, particularly under transient loads typical in AI workloads. Ghaffari et al. [27] emphasized that even in dielectric fluids, boiling behavior and wall temperature response evolve over millisecond timescales and sub-minimeter spatial scales. Their use of dense thermocouple arrays captured rapid bubble formation and collapse, demonstrating the need for instrumentation that can resolve short-lived thermal excursions that is a concern directly relevant to SPIC systems operating near thermal thresholds.

For further highlighting this need of improvement, Zhang et al. [36] investigated single-phase immersion cooling using cold plates with PAO-4 and Noah@3000A. Their findings revealed that cold plate design and fluid properties significantly influence localized thermal resistance and flow distribution—effects that traditional axial measurements may overlook. These insights are increasingly relevant as compact, high-flux electronics like AI chips and advanced packaging architectures demand more precise thermal control. As illustrated in Fig. 6, regions of stagnation and uneven coolant velocity distribution emphasize the importance of geometry-aware design in immersion systems.

Beyond thermal performance, SPIC also aligns well with the economic and deployment requirements of modular data centers, particularly in edge and AI-focused infrastructure. Kanbur et al. [24] demonstrated that SPIC systems can sustain high reliability while achieving lower operating temperatures and reduced energy consumption. These advantages enable plug-and-play deployment models, reduce cooling infrastructure complexity, and contribute to energy-proportional computing—critical traits for sustainable HPC and AI operations.

3. Technological developments and innovations

Building on the aforementioned fundamentals, the next critical determinant of SPIC performance lies in the dielectric fluids themselves, whose thermo-physical and chemical properties directly shape heat

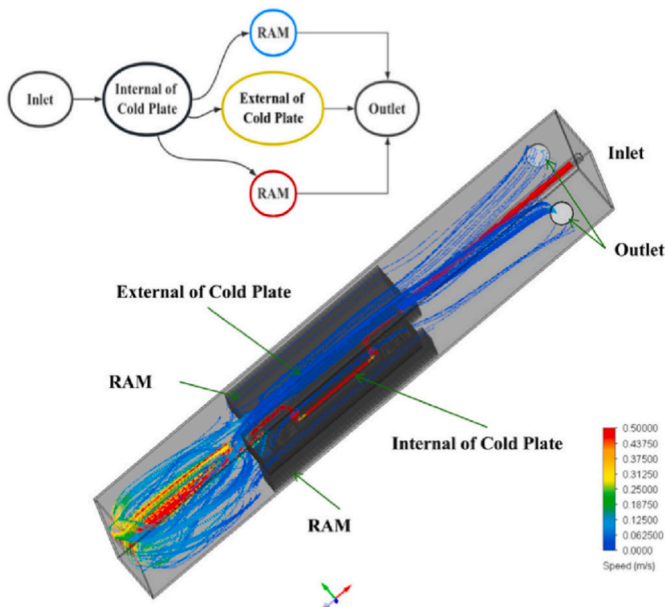


Fig. 6. Flow distribution inside the cold plate under immersion cooling with PAO-4 and Noah@3000A, visualizing areas of high and low velocity. Regions of recirculation and asymmetric flow patterns suggest localized cooling disparities that bulk thermal measurements may fail to capture. Figure adapted from Zhang et al. [36].

transfer, stability, and sustainability outcomes. While this section primarily focuses on SPIC, selected studies on related cooling architectures and diagnostics are referenced where their findings provide transferable insights into materials behavior, flow management, reliability, or system-level constraints relevant to SPIC deployments.

3.1. Coolant materials: advances in dielectric fluids

The performance of single-phase immersion cooling (SPIC) is strongly influenced by the thermophysical properties and chemical stability of the selected coolant. Key parameters include thermal conductivity, specific heat, viscosity, dielectric strength, and long-term compatibility with IT hardware. Advances in coolant development for SPIC systems had focused around three classes of fluids: synthetic hydrocarbons, fluorinated liquids, and nanoparticle-enhanced oils.

3.1.1. Synthetic hydrocarbons and fluorinated liquids

Hydrocarbon-based fluids such as mineral oils and synthetic esters offer moderate thermal conductivity and good material compatibility.

Fluorinated liquids, notably 3M Fluorinert™ (e.g., FC-40) and Novec™, provide high dielectric strength and chemical stability but are more expensive and have lower thermal conductivities.

To illustrate their practical performance, experimental tests by Muneeshwaran et al. [8] using FC-40 in a 1U server tank to demonstrate its ability to maintain thermal stability and deliver effective cooling for GPU-intensive workloads. As shown in Fig. 7(a), increasing the flowrate significantly reduces thermal resistance and temperature differential across various fin pitches, underscoring the need for higher coolant velocities to compensate for FC-40's limited thermal conductivity. Furthermore, Fig. 7(b) illustrates that FC-40 sustains acceptable thermal resistance and temperature margins even under escalating power loads up to 700 W, confirming its robustness in high-performance scenarios.

Beyond data centers, complementary studies by Liu et al. [32] on SPIC fluids for lithium-ion battery modules revealed that fluids with higher thermal conductivities and heat capacities achieved more uniform temperature control and reduced thermal gradients, showing insights that are transferable to data center applications.

White papers by LiquidStack [37] and Bunger and Lin [38] presented comparative property data for fluorinated and hydrocarbon-based fluids. These reports confirmed that while fluorinated fluids are superior in terms of dielectric performance and thermal stability, hydrocarbon-based fluids exhibit better energy efficiency due to lower viscosity and pumping power requirements.

3.1.2. Nanofluid-enhanced dielectrics

The integration of high-thermal-conductivity nanoparticles into base fluids is considered as a new class of nanofluid dielectrics for SPIC systems. Luo et al. [39] reported that SiC nanoparticles dispersed in base oil significantly increased effective thermal conductivity up to ~11.7% over the base fluid and specific heat. Their optimized formulation improved convective heat transfer without compromising dielectric integrity under standard loading conditions.

Early molecular dynamics investigations provided foundational mechanistic insight for this additive. Kang et al. [40] showed that copper-argon nanofluids exhibit a nearly linear increase in thermal conductivity with nanoparticle volume fraction, and attributed this to the formation of a solid-like nanolayer (~0.6 nm) surrounding the nanoparticles. Although this study was not tailored to SPIC fluids, it established the nanolayer effect as a key mechanism underlying nanofluid thermal enhancement.

More recent studies have validated and extended these findings under conditions more relevant to cooling applications. Ghahremanian et al. [41] demonstrated, through MD simulations of two-phase argon-copper nanofluids in rough nanochannels, that nanoparticle-induced interfacial effects dominated over wall roughness in enhancing condensation heat transfer. Similarly, Batool et al. [42] emphasized the role of interfacial nanolayers in improving the effective

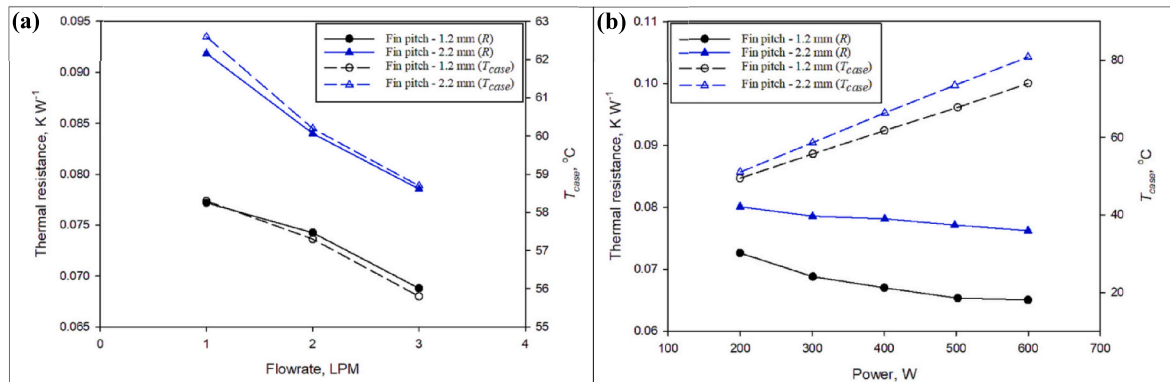


Fig. 7. (a) Thermal resistance and temperature difference versus flowrate for FC-40 across different fin pitches. (b) Performance of single-phase immersion cooling system under varying heat loads using FC-40 (T-configuration, flowrate = 3 LPM, inlet temperature = 35 °C). Figures adapted from Muneeshwaran et al. [8].

conductivity of hybrid nanofluids, confirming that interfacial structuring enhances convective performance. Complementary computational analyses by Faisal et al. [43] further reinforced that nanolayer thickness directly correlates with thermal transport efficiency, as illustrated in Fig. 8, where increasing normalized nanolayer thickness (h) significantly enhances the Nusselt number across the similarity variable (η), highlighting the critical role of interfacial layers in optimizing heat transfer.

While these studies highlight promising nanoscale mechanisms, a comprehensive review by Sun et al. [44] emphasized that dispersion stability, nanoparticle sedimentation, insulation reliability, and long-term degradation remain major barriers to practical deployment of nanofluids in data center SPIC systems.

3.1.3. Material compatibility and system-level considerations

Coolant compatibility with system materials (e.g., gaskets, PCB coatings, cable insulation) is essential for reliability. The Open Compute Project [19] published guidelines detailing material test procedures and compatibility metrics for immersion fluids. Their document emphasizes mechanical integrity, resistance to aging, and chemical inertness under prolonged submersion.

At the system level, coolant selection also affects energy efficiency and design trade-offs. Liu et al. [45] applied a multi-objective optimization approach to assess coolant impact on cabinet-level SPIC systems. Their analysis revealed that fluids with higher thermal conductivity and lower viscosity improved temperature uniformity and lower pump energy consumption.

3.2. System design: technical evolution in tank structures, server layout, coolant circulation, and deployment strategies

Recent developments in single-phase immersion cooling (SPIC) have followed a clear trajectory of technical refinement, transitioning from basic tank immersion concepts to highly modular, data-center-scale deployment systems. This section organizes these innovations into a sequential flow of technical advancement, starting with fundamental tank configurations and progressing toward complex integration and deployment considerations.

3.2.1. Foundational tank design and flow path optimization

Early SPIC system designs emphasized straightforward tank

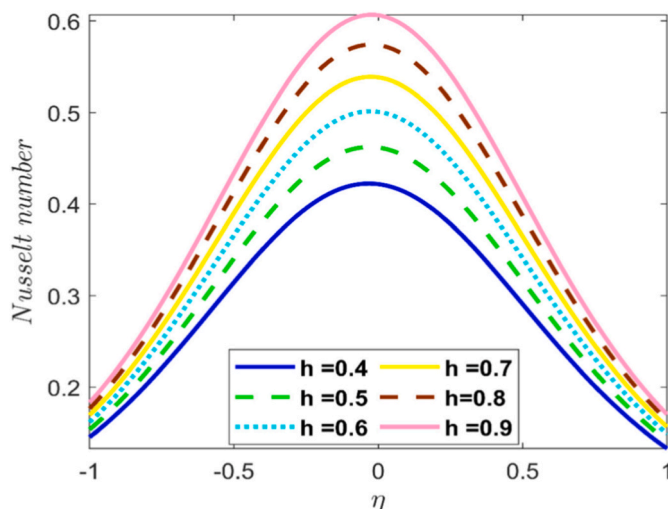


Fig. 8. Variation of the Nusselt number with the similarity variable (η) for different normalized nanolayer thickness values ($h = 0.4\text{--}0.9$). The results demonstrate that increasing nanolayer thickness significantly enhances convective heat transfer, underscoring the pivotal role of interfacial structuring in thermal transport optimization. Figure adapted from Faisal et al. [43].

geometries to ensure full submersion and direct coolant contact with electronic components. Cheng et al. [46] combined experiments with COMSOL simulations using Novec 7100, showing that flow rate and heatsink material choice strongly influenced thermal uniformity and the circulation speed is a key driver for minimizing hot spots and improving energy efficiency.

At rack scale, Kim et al. [14] performed a computational study of an immersion-cooled server stack with FC-40 coolant. They implemented a T-shaped inlet–outlet manifold (three inlets, two outlets), shown in Fig. 9(a), and demonstrated that this immersion design outperformed both air and hybrid cooling configurations by maintaining lower CPU temperatures and stability under high TDP loads.

Complementing these simulations, Li et al. [47] experimentally tested a 16-server tank equipped with distribution manifolds and a porous panel (see Fig. 9(b)). Their results showed that this design is critical in maintaining temperature uniformity with temperature deviation being less than 4.8 °C across GPUs regardless variations in inlet temperature, flow rate, and system load. This confirmed that internal flow control features such as baffles and porous structures are essential to suppress stratification and dead zones.

Scaling up to data-center size, the Baidu–Intel Scorpio 5.0 ICPS project [48] integrated SPIC tanks into production environments, addressing not only heat removal but also rack compatibility, power supply integration, vibration stability, and material compatibility, as depicted in Fig. 9(c). Their evaluation underscored that immersion tanks must be engineered holistically, where flow distribution hardware, system monitoring, and reliability testing converge to ensure safe long-term operation.

Other comparative and numerical studies also reinforce these findings. For example, Rongyang et al. [9] analyzed oil-based SPIC systems, highlighting the influence of local heatsink fin geometry on tank thermal performance, while a comparative single-server study showed that jet-assisted immersion enhances flow mixing compared with pure immersion, reducing localized hotspots. These results support the broader conclusion that tank flow path optimization, circulation strategies, and server-level component design must be co-engineered to achieve stable, uniform thermal performance in SPIC deployments.

3.2.2. Vertical rack integration and expansion systems

Building upon foundational tank principles, large-scale vertical integration has been pursued to accommodate higher densities and ensure data-center compatibility. Chang et al. [49] reported on the development of an advanced 48U single-phase immersion cooling system, shown in Fig. 9(d), designed for commercial deployment. The system incorporated block filling modules to reduce coolant inventory, a flow field uniformity subsystem, and customized heatsinks tailored for immersion operation. Integrated CDU connections and redundant pump configurations provided reliability, while modular expansion allowed scaling without major retrofits. These refinements demonstrate how vertical immersion systems are evolving into enterprise-ready infrastructure solutions.

Similar insights come from Liu and Yu [50], who examined server arrangement and spacing in immersion environments. Their optimization study on two-phase tanks showed that intervals of 15–30 mm between servers strongly affected flow distribution and heat transfer. Although this study focused on boiling systems, the findings emphasize that rack spacing and vertical geometry are critical design parameters for both single- and two-phase immersion deployments.

Together, these studies highlight the trajectory from proof-of-concept vertical racks to high-density, modular, and expandable immersion systems. By integrating flow management subsystems, redundancy features, and careful server arrangement, modern 48U designs are bridging the gap between experimental prototypes and large-scale commercial adoption.

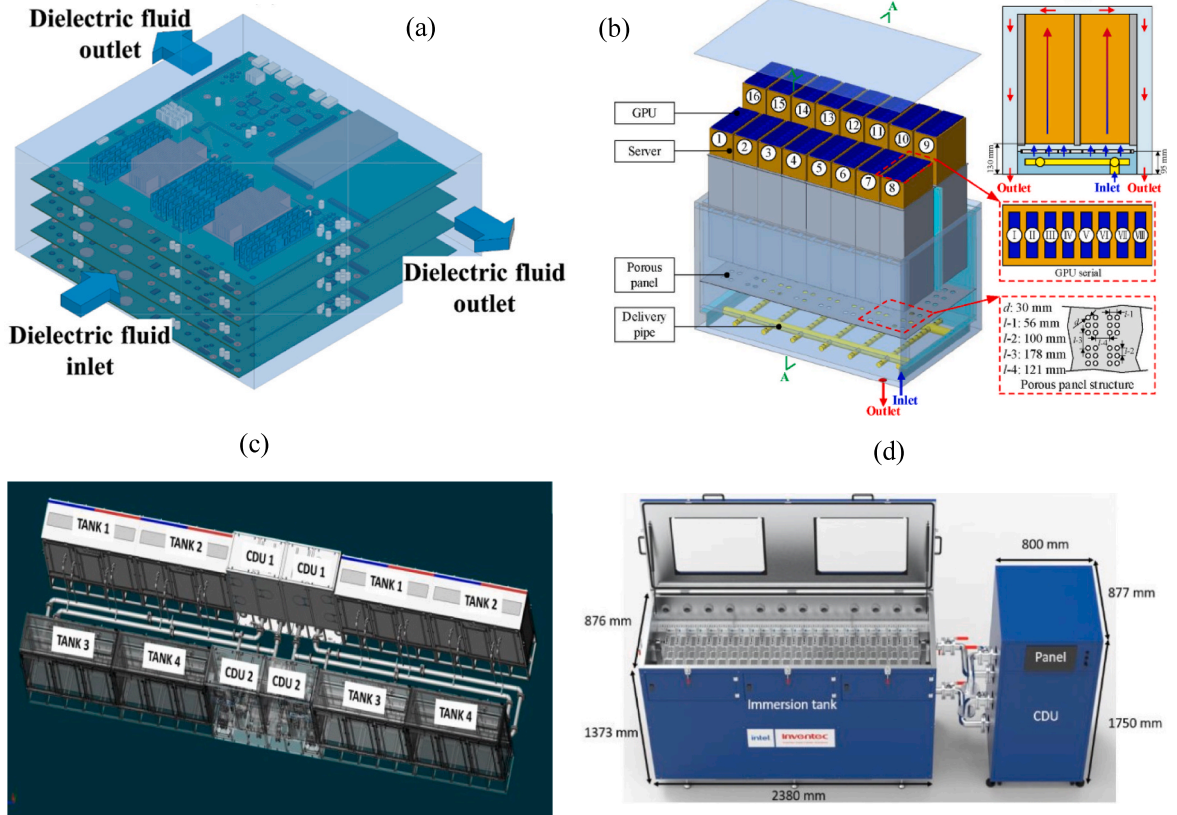


Fig. 9. Representative immersion cooling system configurations across scales and design stages. (a) Computational model of a server stack with a T-shaped manifold (three inlets, two outlets) using FC-40 coolant, reducing CPU temperature and improving thermal stability under high TDP loads (Figure adapted from Kim et al. [14]). (b) Experimental 16-server tank incorporating distribution manifolds and a porous panel, achieving temperature deviation $\leq 4.8^\circ\text{C}$ across GPUs by mitigating stratification and dead zones (Figure adapted from Li et al. [47]). (c) Baidu-Intel Scorpio 5.0 ICRS deployment, addressing rack compatibility, power supply integration, vibration stability, and material compatibility for production environments (Figure adapted from Zheng et al. [48]). (d) Advanced 48U SPIC system for commercial data centers featuring block filling modules, flow field uniformity subsystem, and redundant CDU connections for scalable, reliable deployment (Figure adapted from Chang et al. [51]). These configurations illustrate the evolution from foundational tank geometries to enterprise-scale immersion systems, emphasizing co-engineered flow paths, server layout, and system reliability.

3.2.3. Server orientation and flow symmetry considerations

Optimizing server positioning within immersion tanks emerged as a critical design factor, since orientation influences both convection behavior and flow symmetry. Shrigondekar et al. [25] demonstrated that flow performance can be significantly improved by adopting a T-type inlet-outlet configuration (with inlets positioned centrally and outlets on opposite sides to ensure symmetric distribution) instead of Z-type (diagonal placement) or U-type (inlets and outlets on the same side, which often causes recirculation zones), while also minimizing bypass gaps near heat sinks and incorporating assisted suction to enhance coolant circulation. These combined strategies reduced thermal resistance and lowered case temperatures, underscoring the importance of engineered flow paths that align naturally with convective currents.

Direct evidence of orientation effects was provided by Saini et al. [52], who numerically analyzed mixed convection in an immersion-cooled dual-CPU server using EC-100 fluid. They found that in vertical orientation, where forced flow aligns with buoyancy, heat transfer improved significantly at low Reynolds numbers due to buoyancy-driven secondary flows, resulting in lower maximum component temperatures and more uniform heat removal. In contrast, horizontal orientation yielded localized high-temperature regions at higher flow rates as buoyancy effects weakened, confirming that vertical alignment better sustains thermal performance in SPIC systems under realistic low-flow conditions.

Overall, these findings highlight the critical role of server orientation and flow symmetry in SPIC performance. By improving flow distribution

and reducing temperature non-uniformities, these strategies help prevent localized overheating and enhance reliability in high-density deployments.

3.2.4. Coolant circulation enhancement through targeted flow

With tank layout and server positioning being addressed, the focus shifted to active circulation strategies that move beyond purely passive convection. Several studies highlight the importance of guided or controlled coolant flow in improving thermal uniformity and reducing hotspot risks.

Natural convection in SPIC systems is inherently constrained by weak buoyancy-driven flow, leading to long stabilization times, non-uniform temperature distribution, and insufficient heat transfer capacity for high-power-density hardware. Huang et al. [53] compared buoyancy-driven and pump-driven SPIC systems, showing that pump-driven configurations significantly outperformed natural convection by reducing average coolant temperature by 55.5%. The study also noted that increasing flow rate enhances convective heat transfer and temperature uniformity but introduces higher flow resistance, highlighting the trade-offs in circulation design. Similarly, Cheng et al. [46] experimentally and numerically evaluated a Novec 7100-based SPIC system under forced circulation and demonstrated that increasing flow velocity from 0.4 to 1.2 m/s significantly improved thermal uniformity and reduced stabilization time (from about 6.5 min to 1.5 min), addressing key challenges typically associated with natural convection-driven cooling.

Shahi et al. [54] designed and validated a dynamic flow control device (FCD) for liquid-cooled servers, featuring a V-cut ball valve actuated by a microservo motor to modulate flow rates based on real-time thermal demand. Experimental results confirmed the device could vary flow from 0.09 to 4 LPM, enabling targeted coolant delivery and reducing unnecessary pumping power. These findings highlight that workload-responsive flow steering offers significant efficiency gains compared to conventional constant-flow strategies. These findings highlight that workload-responsive flow steering offers significant efficiency gains compared to conventional constant-flow strategies. Collectively, such flow-routing strategies, whether through manifold design, guided nozzles, or controlled outlets, are critical enablers for achieving consistent immersion cooling performance in larger form factors.

3.2.5. High-density modular deployment and scalability

As single-phase immersion cooling (SPIC) matured, attention shifted toward modularity and scalability for high-density deployment. Zou et al. [55] demonstrated through multi-objective optimization that structural refinements in immersion tanks can simultaneously improve thermal management and reduce coolant consumption. Their study reported reductions of 3.39% in average chip temperature, 3.37% in maximum chip temperature, 1.05% in accessory component temperature, and 36.30% in coolant filling ratio, highlighting that the importance in scaling up the immersion cooling which requires not only higher density but also more efficient use of coolant resources. The optimization analysis identified filler spacing (FILL) and hydraulic radius of the liquid distribution orifice plate (R) as the most influential factors for both thermal performance and coolant reduction, suggesting that precise control of internal tank geometry is critical for scalable SPIC design.

At the same time, Vora [56] emphasized that the rapid growth of workloads such as artificial intelligence, machine learning, and high-performance computing presents significant challenges to existing cooling methods. The study highlights the increasing importance of liquid and immersion cooling as strategies capable of supporting next-generation data center demands, particularly in terms of energy efficiency, adaptability, and scalability.

Khoshvaght-Aliabadi et al. [2] investigated the impact of coolant distribution design on server-level thermal management. By comparing series, parallel, and hybrid cold plate configurations in multi-CPU servers, their results showed that optimized coolant pathways improve temperature uniformity and reduce pumping power requirements, thereby supporting the reliable and scalable deployment of liquid cooling in high-density data centers.

The industrial validation was carried out by Zheng et al. [48] who examined a single-phase immersion cooling solution developed through collaboration between Baidu and Intel. The resulting Scorpio 5.0 Immersion Cooling Rack Solution (ICRS) demonstrated measurable benefits at the data center scale, including PUE values below 1.1, 7–17% reductions in total cost of ownership over five years, improved CPU performance by 7.51%, and server-level power savings exceeding 30 W. These findings confirm that immersion cooling can deliver both high density and operational efficiency in practice.

3.2.6. Deployment strategy, retrofit challenges, and future trends

At the device level, the OCP OAI System Liquid Cooling Guidelines identify processor modules drawing as much as ~700 W (PCIe) to ~1000 W (OAM), motivating the demand of immersion with the cold-plate [57]. Facility-level performance data further demonstrate the benefits of immersion: reported deployments achieve power usage effectiveness (PUE) values in the range of 1.02–1.04, highlighting substantial efficiency gains over conventional air-based systems [58]. Complementary evidence from recent vendor case studies confirms that the modular and containerized immersion deployments achieve PUE values near 1.12, and can reach in 1.07 in favorable conditions. These

systems also demonstrate the potential for hot-water reuse at ~65 °C, underscoring immersion's contribution to both efficiency and energy recovery [59].

Despite these advantages, retrofit feasibility remains a barrier. Analyses emphasize that transforming existing air-cooled facilities into immersion-ready sites entails major complexity in manufacturing definitions, transport readiness, data center integration, and commissioning, making such projects costly and operationally demanding [60]. Instead, purpose-built immersion facilities are often identified as more practical for adoption, as they can be designed around liquid logistics from the outset. Early modular design concepts already highlighted the importance of standardized distribution networks, manifold arrangements, and serviceability protocols to ensure scalable operation [59]. These priorities remain central in more recent integration frameworks, which expand to commissioning logistics and operational safety considerations [60].

Serviceability expectations also require redefinition in liquid environments. The OCP OAI guidelines explicitly note that it is impractical to expect liquid-cooled systems to be as easily serviced as air-cooled counterparts; however, they recommend design provisions such as modular field-replaceable units and internal quick-connects to simplify component replacement and minimize downtime [57].

Looking ahead, industry and academic sources converge on the view that immersion cooling, alongside cold-plate approaches, will continue to expand as workloads in AI and high-performance computing intensify. This evolution is not only driven by efficiency requirements but also by opportunities for heat recovery and sustainability integration, as noted in both academic reviews and vendor reports [6,29,59]. The long-standing recognition that cooling can account for up to 40% of total data center energy demand underscores the importance of these liquid-based strategies in achieving future efficiency and environmental goals [61].

3.3. Thermal management enhancements

Recent advancements in single-phase immersion cooling (SPIC) for data centers have increasingly leveraged artificial intelligence (AI) and machine learning (ML) to improve coolant flow control, enable real-time diagnostics, and provide accurate automated thermal predictions. These developments focus on reducing energy consumption, increasing thermal reliability, and maintaining optimal operating conditions through predictive and adaptive control strategies.

One emerging research direction involves reinforcement learning (RL) frameworks for dynamic cooling optimization. Deep RL has been applied to real-world data center cooling systems to learn optimal control policies that reduce cooling energy use while maintaining target temperatures under varying load conditions [62]. Context-aware RL further integrates thermal storage system behavior, as illustrated in Fig. 10(a), enabling the cooling plant to adapt to temporal variations in energy pricing and workload profiles [63]. The RL framework proposed by Zhan et al. [64] integrates four key components: a server room layout, a topological dependency graph, a T-symmetry enforced thermal dynamics model (TTDM), and a sample-efficient offline policy optimization pipeline. These modules work together to provide physics-informed, adaptive cooling control, as shown in Fig. 10(b). In addition, RL approaches have been applied in commercial cooling systems, demonstrating the feasibility of deployment within existing infrastructure [39]. These developments align with insights from recent systematic reviews of RL applications for data center energy efficiency, which highlight the growing maturity of such methods and their integration with emerging thermal management strategies [65].

Beyond RL-based optimization, predictive modelling approaches such as digital twin (DT) frameworks have emerged for advanced thermal management in SPIC systems. DT frameworks integrate real-time monitoring with predictive modelling to create a virtual representation of the thermal status of data center [66]. Predictive DTs combine

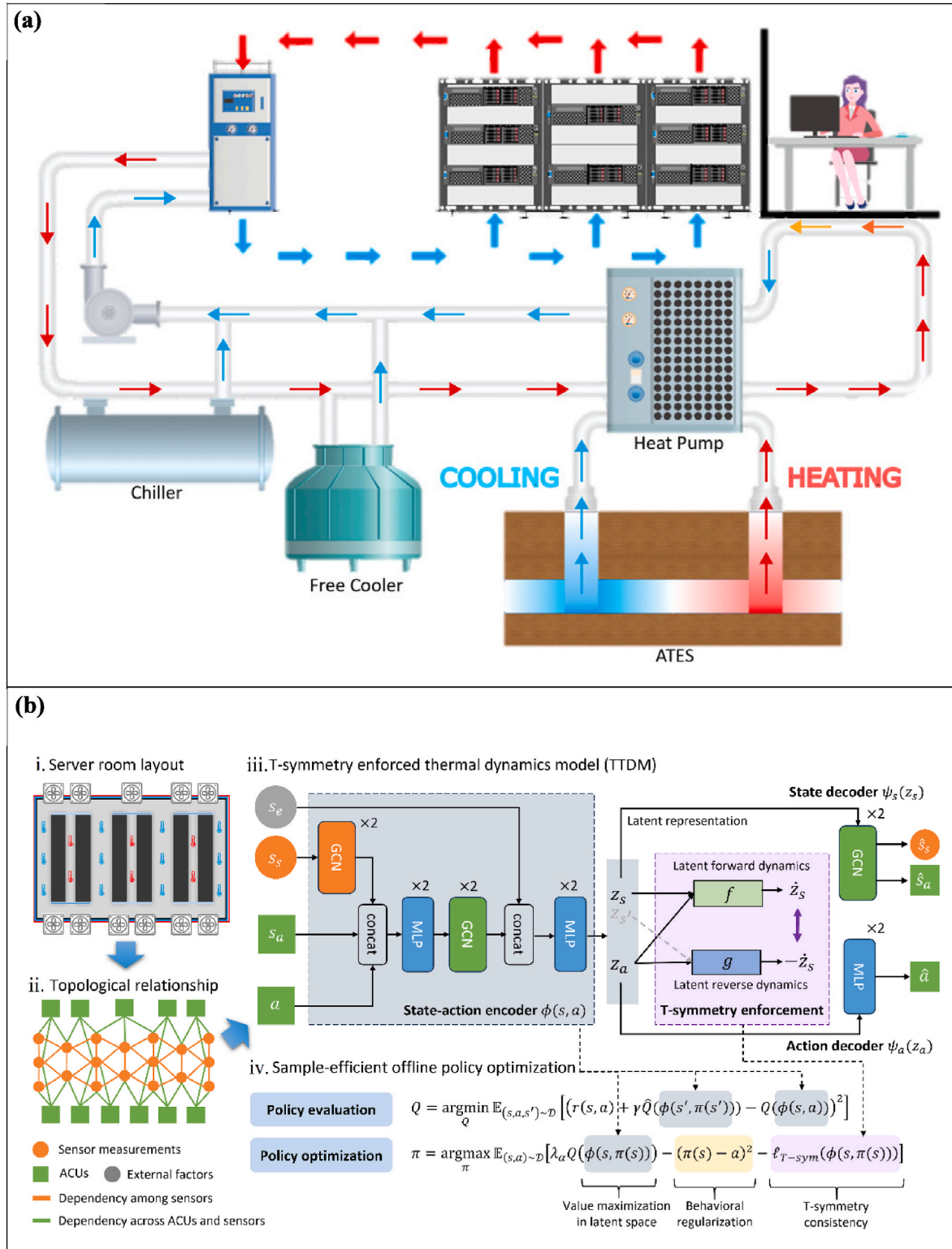


Fig. 10. Integration of hybrid cooling and reinforcement learning-based optimization strategies for advanced thermal management in data centers. (a) Aquifer Thermal Energy Storage (ATES) system coupled with an electric chiller, free cooler, and heat pump for combined comfort heating and server cooling, demonstrating seasonal groundwater-based energy storage for efficiency gains (Figure adapted from Leindals et al. [63]). (b) Physics-informed offline reinforcement learning (RL) control framework integrating four key components: (i) Server room layout representing real-world cooling system configuration; (ii) Topological dependency graph encoding actuator-sensor relationships for structured state representation; (iii) T-symmetry enforced thermal dynamics model (TTDM) ensuring physics consistency and robust generalization; (iv) Sample-efficient offline policy optimization pipeline leveraging historical data and behavioral regularization to reduce cooling energy while maintaining thermal safety (Figure adapted from Zhan et al. [64]).

ML algorithms with reduced-order models to enable proactive cooling control and fault prevention [67]. These systems can forecast temperature evolution under different operational scenarios, allowing for pre-emptive adjustments to flow rates and coolant distribution.

Real-time system diagnostics have been enhanced through gray-box modelling and state-space modelling approaches, addressing the difficulty of modelling highly transient thermal behavior under fluctuating workloads. Gray-box transient temperature models integrate physical laws with data-driven components to provide accurate temperature evolution predictions during rapid load changes [68]. Similarly, state-space models with online parameter identification have achieved high prediction accuracy in rack-based cooling systems, enabling adaptive recalibration under changing conditions [69]. Such methods support early detection of abnormal cooling behavior, reducing downtime risks.

Fault detection mechanisms now incorporate energy monitoring and temperature analysis to identify cooling inefficiencies and potential system failures before they escalate [70]. By embedding diagnostic intelligence into the control loop, operators can ensure continued cooling performance without compromising energy efficiency.

The integration of these AI-assisted flow control, real-time diagnostics, and predictive thermal modelling capabilities into SPIC represents a shift from reactive to proactive cooling management. This not only improves thermal stability but also enhances energy efficiency, aligning with green computing goals, which can translate into substantial reductions in PUE and operational costs for hyperscale data centers [71].

Together, these innovations establish the technological foundation upon which SPIC is being deployed in practice, particularly within data centers and HPC environments, where demands for efficiency, scalability, and sustainability are most acute.

Beyond passive bath-type immersion, jet-enhanced immersion cooling has emerged as an effective strategy to further augment convective heat transfer in SPIC by actively disrupting thermal boundary layers and enhancing local fluid renewal, consistent with established jet-impingement heat transfer principles [72].

Experimental studies directly targeting data-center-scale systems have demonstrated that submerged jet impingement can significantly reduce component peak temperatures and improve spatial thermal uniformity compared with conventional immersion cooling. For example, Huang et al. [73] reported maximum local temperature reductions of up to $\sim 20^\circ\text{C}$ and substantial hotspot suppression in jet-enhanced SPIC configurations under data-center-relevant power densities. Complementary investigations further showed that optimized jet arrangements and flow rates can achieve surface temperature non-uniformities on the order of 1°C or lower, which is critical for mitigating thermally induced stress in high-power electronics [74]. Recent advances further integrate data-driven optimization, where machine-learning-assisted jet control enables adaptive flow distribution to balance cooling performance and pumping power in response to dynamic workloads [74]. Collectively, these developments position jet-enhanced immersion cooling as a scalable and energy-efficient extension of SPIC for high-power AI and HPC applications, consistent with broader trends in next-generation electronics and data-center thermal management [75].

4. Applications in data centers and HPC systems

SPIC has progressed beyond laboratory-scale validation toward deployment-relevant demonstrations in data-center and high-performance computing environments. Recent peer-reviewed studies report experimental and operational evidence of SPIC applied to high-density computing systems under realistic thermal and power conditions.

Sun et al. [76] conducted an experimental investigation of a single-phase immersion cooling unit designed for high-density

computing power data-center applications. Their results demonstrated stable long-term operation, effective suppression of component-level temperature rise, and improved thermal uniformity under elevated heat-flux conditions representative of modern computing workloads. The study confirms the suitability of SPIC for sustained operation in dense computing environments where air cooling approaches thermal limits.

Complementary operational evidence is provided by Misra et al. [77], who examined immersion-cooled servers operated beyond nominal frequency and power limits in a production-adjacent data-center setting. Their findings show that SPIC enables increased thermal headroom, supports higher power densities, and maintains system reliability under overclocked conditions. These results highlight the potential of SPIC to enhance performance margins and operational robustness in HPC-oriented infrastructures.

Collectively, these studies demonstrate that single-phase immersion cooling can deliver tangible thermal and operational benefits in data-center-relevant deployments, supporting its continued evaluation and adoption for high-density AI and HPC systems.

4.1. Hyperscale and containerized data center deployments

Scalability remains a critical design consideration for deploying SPIC in modern data centers, where the transition from single-rack setups to containerized or hyperscale installations introduces unique thermal, mechanical, and operational challenges. While early SPIC deployments were limited to single-server or small-cluster systems, recent advancements have enabled high-power-density, multi-rack 48U configurations capable of supporting dense commercial workloads [49]. Achieving scalability requires careful coordination of tank volume, heat exchanger capacity, and fluid circulation pathways to maintain uniform thermal performance and allow for modular expansion.

When scaling SPIC to containerized or hyperscale environments, maintaining thermal uniformity across high-power-density racks becomes increasingly challenging. Computational studies highlight that manifold design and inlet-outlet placement strongly influence rack-scale temperature gradients, while targeted flow enhancements such as localized jets can mitigate hotspots around non-uniform thermal loads [14,52].

David et al. [78] conducted simulation-based evaluations comparing a single-server setup with a multi-server cluster in a shared immersion tank, focusing on a 2.66 kW 2U server representative of HPC workloads. Results show that while multi-server configurations can deliver high cooling efficiency, balanced flow distribution is critical to prevent uneven cooling and hotspots as system complexity increases. The study also assessed the impact of server removal for maintenance on cluster cooling performance, underscoring an operational challenge for large-scale SPIC deployments.

At the industry level, formalized guidelines and technical reviews provide critical frameworks for immersion cooling adoption. Open Compute Project [79] outline system-level design constraints, safety requirements, and interoperability provisions for immersion deployments, including leak detection, fluid containment, and maintenance protocols. Open Compute Project [80] provide detailed recommendations for hardware adaptation, covering material compatibility, connector sealing, and firmware adjustments for immersion environments. Complementing these, Open Compute Project [81] defines standardized thermal and electrical performance metrics, including viscosity, dielectric properties, and environmental compliance parameters, to ensure cross-vendor compatibility and operational safety. In parallel, technical assessments reinforce the necessity of such standards for hyperscale adoption. Robinson [82] highlights operational and environmental drivers behind immersion cooling, emphasizing interoperability and lifecycle considerations in large-scale data centers. Similarly, Azarifar et al. [83] and Zhou et al. [6] identify material compatibility, fluid stability, and modular deployment as critical

challenges that require unified specifications to maintain reliability and sustainability across global infrastructures.

These combined frameworks and analyses are essential for hyper-scale deployments, where consistent implementation of fluid standards, hardware integration, and environmental safeguards enables scalable operations while minimizing total cost of ownership and compliance risks.

4.2. Cloud and AI-oriented data centers: energy efficiency and sustainability

Energy efficiency remains a critical performance metric for data centers, particularly as AI and cloud computing workloads drive unprecedented infrastructure demand. Power Usage Effectiveness (PUE), defined as the ratio of total facility energy consumption to IT equipment energy use, has become the industry's primary benchmark for operational efficiency. Lower PUE values signify improved performance by reducing non-IT energy overhead, which translates directly into lower operating costs and reduced greenhouse gas emissions. Current estimates place typical PUE values for large U.S. data centers in the range of 1.4–1.6, while advanced liquid cooling systems offer pathways toward values approaching 1.2. These improvements are especially significant for hyperscale and AI-intensive facilities, where cooling energy accounts for a substantial portion of overall energy use [84].

Recent studies had highlighted the potential of SPIC to significantly improve PUE by reducing cooling infrastructure power consumption and supporting higher rack densities without overheating risk. Huang et al. [53] demonstrated that replacing buoyancy-driven circulation with pump-driven configurations lowered coolant temperatures by 55.5% and improved PUE by 11.6%, underscoring the impact of circulation strategies on energy efficiency. Similarly, Hnayno et al. [7] showed that optimized fluid flow paths in advanced liquid cooling topologies enhance heat dissipation per unit pumping energy, translating to measurable PUE improvements in practical deployments.

For high-performance workloads such as AI training clusters, Ma et al. [17] proposed an AI-enabled predictive energy saving planning framework that combines Transformer and GRU networks to dynamically forecast coolant return temperatures under variable load conditions. As shown in Fig. 11, the model architecture integrates a Transformer layer for feature extraction with GRU layers for temporal sequence prediction, enabling proactive adjustments of cooling system parameters to maintain thermal stability and optimize energy efficiency. This predictive approach reduces unnecessary pumping energy and supports real-time operational decisions, which are critical for data centers hosting heterogeneous computing hardware with highly fluctuating heat generation profiles.

From a sustainability perspective, life cycle assessment (LCA)-based evaluations [4] had revealed that immersion cooling can substantially reduce carbon emissions compared to air cooling, not only by improving operational energy efficiency but also by lowering embodied energy through reduced use of traditional air-handling units. The study emphasized that cooling technology selection plays a major role in the overall environmental footprint of data center operations.

At the component level, FC-40 dielectric fluid performance analysis [8] quantified both thermal performance and pumping energy requirements, indicating that dielectric fluid choice impacts not only heat transfer but also system-wide electrical consumption. Complementing these findings, numerical studies on mixed convection [52] identified design configurations that exploit buoyancy-driven flow to supplement forced convection, thereby reducing pump power and enhancing energy efficiency in dense rack arrangements.

Energy efficiency improvements in immersion-cooled data centers are driven by three interconnected factors. Firstly, thermofluidic optimization of system design ensures effective heat removal while minimizing hydraulic losses. Secondly, predictive control and load management enable proactive adjustment of operating conditions to

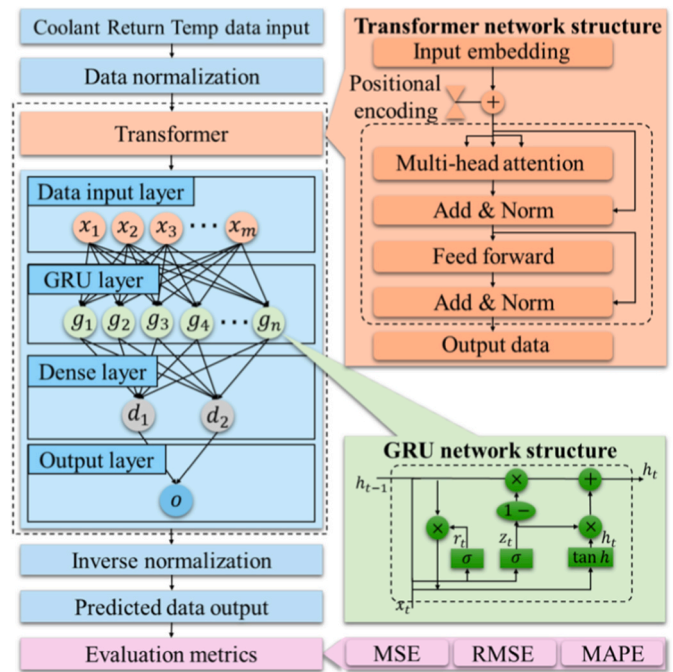


Fig. 11. Network architecture of the Transformer-GRU model for AI-enabled predictive energy saving planning in liquid-cooled data centers. The framework combines Transformer layers for feature extraction and GRU layers for temporal sequence prediction, supporting proactive thermal management. Figure adapted from Ma et al. [17].

match dynamic workload demands, thereby reducing unnecessary energy consumption. Thirdly, environmentally conscious selection of fluids and components reduces both operational and embodied energy while maintaining long-term reliability. Collectively, these strategies are particularly critical for AI and cloud infrastructures, where computational intensity and workload variability require adaptive, high-efficiency cooling solutions.

4.3. Accelerated computing platforms: GPU/TPU clusters and edge AI

The rapid expansion of large-scale AI workloads, including large language models (LLMs), vision-language models (VLMs), and edge inference engines, has driven an unprecedented rise in rack-level power densities. In GPU/TPU clusters, sustained training and inference loads often push thermal envelopes beyond the limits of conventional air cooling, making immersion cooling an increasingly attractive alternative.

Zhu and Wang [85] introduced an immersion-cooling-aware scheduling algorithm that reduced energy consumption by up to 53.2% and shortened training delays by 22.5% for LLM workloads on GPU clusters, underscoring immersion cooling's potential to enable energy-efficient and performance-optimized AI computing at scale. In addition, Bansode et al. [86] conducted 144-h elevated-temperature tests over an ambient range of 25 °C–55 °C, confirming that single-phase immersion cooling maintains thermal stability and system reliability during prolonged operation without mechanical failures.

The deployment of high-power thermal test vehicles (TTVs) has become essential for validating thermal strategies under extreme loads. For instance, Liu et al. [87] developed a TTV equipped with a two-phase cooling system to emulate next-generation AI accelerators and edge compute modules, demonstrating heat dissipation capabilities exceeding 1 kW per device. Although focused on two-phase cooling, this work underscores the importance of early-stage validation platforms, which are equally critical for SPIC systems expected to accommodate similar heat flux levels in future AI clusters.

Scalable AI clusters also require interconnect solutions capable of supporting ultra-high bandwidth with minimal power overhead. Kurata et al. [88] demonstrated that immersion-compatible silicon photonic transceivers maintain stable optical performance during continuous operation in both HPC and AI environments. Fig. 12 presents eye diagrams for transceivers tested under four conditions: air baseline, single-phase immersion (Coolant A), and dual-phase immersion (Coolants B and C) at both room temperature and boiling points. The results reveal negligible signal degradation for Coolants A and B, both with low dielectric constants, while Coolant C (dielectric constant = 7.5) exhibited noticeable eye closure and increased bit error rate (BER) at elevated temperatures. These findings highlight the critical role of dielectric property optimization in preserving signal integrity for high-speed optical links in immersion environments.

Looking forward, future interconnect technologies leveraging highly parallel silicon photonic links driven by integrated Kerr frequency combs are projected to scale to multi-terabit and even petabit-level throughput while achieving sub-pJ/bit energy efficiency, addressing both bandwidth and power constraints in distributed AI and HPC systems [89].

Finally, predictive analytics are being integrated into immersion-cooled AI clusters to enable proactive thermal management. Pérez et al. [90] demonstrated that a GRU-based recurrent neural network could forecast coolant-side thermal conditions 1 min in advance with a mean absolute error of approximately 0.75 °C and an R^2 value of 89.5%, providing the capability to adjust cooling strategies before critical thresholds are reached. Building on this, Pratheek et al. [91] developed an attention-enhanced LSTM encoder-decoder model with entropy minimization, achieving up to 22%, 15%, and 20% improvements in prediction accuracy for 10s, 30s, and 60s horizons, respectively. These approaches illustrate how machine learning-driven prediction frameworks can support Dynamic Thermal Management Systems (DTMS),

reducing unnecessary pumping power, enhancing reliability, and contributing to overall energy savings in single-phase immersion cooling deployments.

4.4. Pilot-scale and production-adjacent deployments

Pilot-scale and production-adjacent deployments of SPIC, as reported in the peer-reviewed literature, had provided critical insights into real-world thermal stability and operational reliability. A representative hybrid architecture is shown in Fig. 13, which integrates immersion cooling with a serpentine heat exchanger to enable high-temperature water loops (up to 45 °C) and heat recovery capability. This configuration illustrates how modern SPIC systems combine tank-based immersion with facility water circuits for efficient heat removal. A six-month pilot deployment reported by Wang et al. [23] demonstrated that CPU and other key component temperatures remained stable with no observable performance degradation when fluid quality and material compatibility were properly managed. Periodic analysis of fluid properties, including acidity, water content, viscosity, and dielectric strength, confirmed minimal variation over time, while material extraction tests showed negligible impact on weight and appearance after prolonged immersion. These results suggest that, under controlled conditions, SPIC can achieve reliability levels comparable to air-cooled systems even under high-density, continuous workloads, paving the way for broader adoption in hyperscale data centers.

Comparative evaluations by Sivaraju et al. [92] analyzed the thermal performance of single-phase immersion cooling (SPIC) for a two-socket server in tank-based systems featuring large, open-bath designs and sled-based systems employing individual, rack-mountable enclosures, using computational fluid dynamics (CFD) simulations validated with experimental measurements. Results showed that both configurations maintain CPU temperatures below 70 °C across a range of inlet

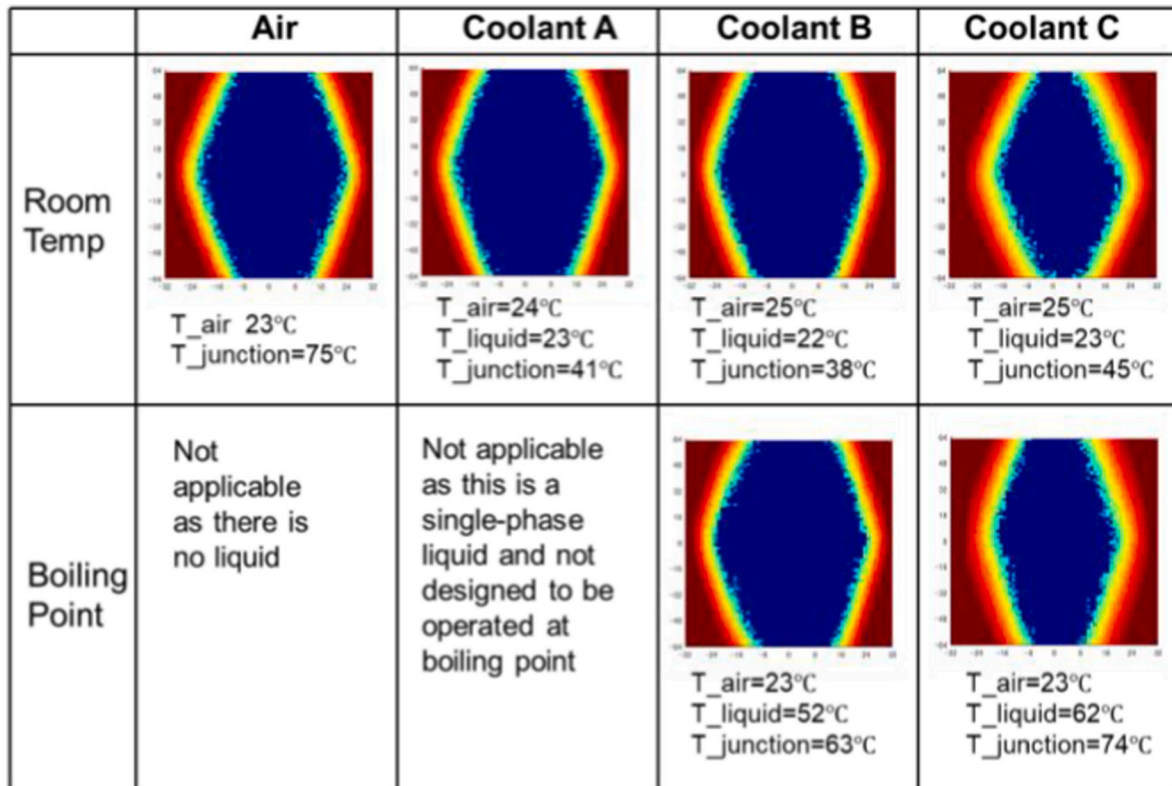


Fig. 12. Silicon photonic transceivers exhibit stable signal integrity under multiple immersion cooling conditions, underscoring their suitability for reliable, high-bandwidth interconnects in AI and HPC environments. Figure adapted from Kurata et al. [88]. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

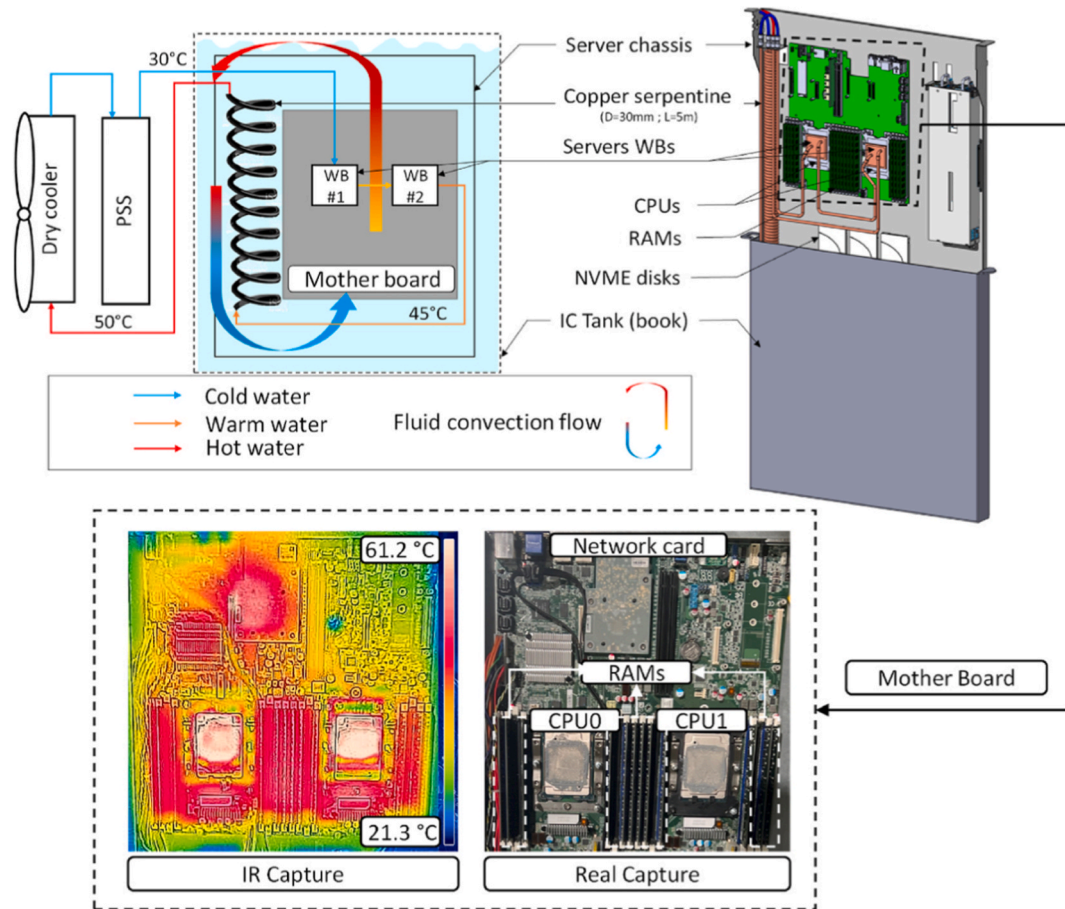


Fig. 13. Architecture and thermal analysis of the hybrid single-phase immersion/liquid cooling system developed by Hnayno et al. [93]. The schematic illustrates the system layout integrating direct-to-chip water blocks with a natural convection immersion tank and a serpentine coil for heat exchange with facility water. The lower panel shows (left) an infrared thermal image indicating hotspot distribution across server components (CPUs, RAMs, and network card) and (right) the corresponding real image for reference. The system enables high inlet water temperatures (up to 45 °C) while maintaining component reliability, facilitating energy savings and potential heat recovery. Figure adapted from Hnayno et al. [93].

temperatures and flow rates, indicating negligible differences in thermal performance. However, the study highlighted trade-offs in mechanical design and operational logistics: while tank-based systems provide conventional immersion layouts, they require full-scale infrastructural modifications, whereas sled-based configurations facilitate retrofits with minimal changes to existing rack architectures, improving serviceability and deployment flexibility. At a broader scale, quantitative evaluations demonstrate that immersion cooling can reduce PUE to 1.02–1.04 and enable rack power densities approaching 250 kW, significantly lowering energy use and floor space requirements compared to air-cooled systems [58]. These advancements support high-performance computing workloads and address the inherent limitations of airflow-based cooling in hyperscale environments.

Laboratory-scale and production-ready demonstrations have extended these findings to high-power server platforms. Immersion cooling of multi-kilowatt compute nodes consistently maintained junction temperatures well below throttling thresholds, even under elevated ambient conditions, while eliminating the need for high-speed server fans [86,93]. Measurements of servers submerged in dielectric fluids such as FC-40 exhibited predictable heat dissipation profiles and repeatable thermal behavior across a range of load conditions [8].

Beyond direct thermal performance, immersion deployments have demonstrated significant facility-level efficiency gains. By reducing or eliminating mechanical chiller usage and leveraging opportunities for heat recovery, SPIC and two-phase immersion systems have achieved substantial reductions in total power consumption within high-density

compute environments [93,94]. In addition, immersion cooling enables sustained operation at rack power densities far exceeding those feasible for air-cooled architectures, with reported capacities reaching up to 250 kW per rack [58].

Material compatibility has emerged as a decisive factor in long-term reliability, as continuous exposure to dielectric fluids can impact polymers, elastomers, and printed circuit board coatings. To address these risks, industry guidelines such as those from the Open Compute Project Advanced Cooling Facilities (OCP ACF) initiative provide validated material lists and reference designs to ensure mechanical and chemical stability [19,95]. At the component level, device-integrated heat sinks and multi-parameter-optimized fin structures had demonstrated measurable improvements in heat transfer coefficients without increasing system footprint, supporting efficient natural convection in immersion environments [96,97].

Advanced fluid formulations and thermal interface enhancements further define the trajectory of next-generation immersion systems. Numerical analyses showed that the incorporation of Al_2O_3 nanoparticles into mineral oil significantly improves thermal conductivity and local heat transfer performance, though challenges related to dispersion stability and filtration persist [98]. Complementary to these developments, investigations into PCIe signal integrity have underscored the impact of immersion environments on emerging high-speed interconnects, with stricter thermal and electrical compliance requirements anticipated for PCIe Gen6/Gen7 and beyond [99].

Collectively, these pilot-scale and production-adjacent

implementations establish SPIC as a technically mature and operationally viable cooling strategy for next-generation data centers, particularly when deployed in conjunction with standardized design practices, validated material sets, and advanced thermal optimization methodologies.

5. Reliability and maintenance considerations

Beyond thermal performance, long-term deployment of SPIC in data centers and HPC environments critically depends on material compatibility, system-level reliability, and the integrity of electrical and optical signaling under continuous immersion exposure. In addition, the electrically insulating nature of dielectric immersion coolants enhances human safety by reducing electric shock risk during installation, operation, and maintenance of IT equipment.

The preceding section highlighted how SPIC has been deployed across HPC and data center infrastructures, with particular emphasis on scalability, energy efficiency, and its role in supporting AI and accelerated workloads. While these applications demonstrate the promise of immersion technologies, they also underscore the diversity of system configurations and operational conditions under which SPIC is implemented. To move from individual case studies toward broadly applicable design principles, a comparative evaluation is needed. This section therefore reviews experimental and industry findings on coolant performance, system reliability, and integration strategies, distilling these into best practices that guide the selection of fluids, architectures, and control approaches for different deployment contexts.

Accordingly, the following subsections synthesize experimental evidence and industry practice to address component longevity, coolant ageing, maintenance diagnostics, and signal integrity under immersion conditions.

5.1. Component longevity (servers, storage, AI accelerators)

The operational longevity of servers, storage devices, and AI accelerators in SPIC environments is influenced by a combination of thermal, mechanical, and chemical factors that differ significantly from air-cooled systems. Unlike conventional cooling methods, SPIC eliminates airflow-induced temperature gradients, reduces thermal cycling, and minimizes particulate contamination. These characteristics mitigate fatigue-driven failure modes such as solder joint cracking and PCB warpage, supporting improved overall reliability [23,86].

Controlled experiments confirm that SPIC can maintain thermal stability over extended durations under elevated ambient conditions. Bansode et al. [86] demonstrated consistent junction temperature during a 144-h thermal stress test, conducted across an ambient range of

25 °C–55 °C, with no evidence of thermal throttling or instability. Similarly, a six-month operational study by Wang et al. [23] reported negligible variations in dielectric fluid properties—including acidity, moisture content, viscosity, and dielectric strength—through periodic monitoring. Material extraction tests revealed no significant mass loss or observable degradation in polymers, elastomers, or PCB coatings, indicating strong chemical stability of system components during immersion.

Substrate structural integrity has also been assessed under prolonged liquid exposure. As shown in Fig. 14, cross-sectional analysis of ball grid array (BGA) packages from air-cooled and oil-immersed servers revealed comparable substrate thickness ($\sim 62\ \mu\text{m}$) without signs of delamination or distortion, confirming that immersion does not inherently compromise substrate geometry under controlled conditions [20].

Despite these advantages, immersion introduces unique reliability considerations linked to continuous liquid contact. Shah et al. [21] documented measurable shifts in the electrical characteristics of passive components such as inductors, capacitors, and connectors during extended immersion operation. Variations in inductance, capacitance, and contact resistance were primarily associated with dielectric absorption and partial coolant ingress into insulation layers, emphasizing the need for robust sealing and insulation strategies. Semiconductor devices, while generally stable in immersion environments, exhibit sensitivity to electrical stress and fluid purity. Min et al. [100] reported that leakage current increased significantly under high-voltage bias when trace conductive contaminants were present in the dielectric fluid. Fig. 15 illustrates leakage current increase rates at 3.0 V and 3.1 V for devices cooled in air versus immersion, along with schematic representations of leakage mechanisms in both environments. These results indicate that immersion mitigates, but does not eliminate, leakage under combined electrical and thermal stress, underscoring the importance of stringent fluid purity control and bias management.

Printed circuit boards (PCBs) present additional long-term challenges when immersed at elevated temperatures. Ramdas et al. [22] observed a reduction in elastic modulus for FR-4 substrates following aging, which can influence dimensional stability under mechanical load. Complementary studies by Bansode [101] on halogen-free IS550H substrates reported further reductions in modulus after 720-h immersion in EC100 and PAO6 fluids at 85 °C and 125 °C, suggesting changes in interlaminar adhesion and increased susceptibility to CTE-driven stress during aging. These effects are particularly relevant in high-density systems where localized heat flux can exacerbate thermal gradients if coolant flow is non-uniform.

Reliability evaluations had also been extended to optoelectronic components. Kurata et al. [88] demonstrated that silicon photonics transceivers submerged in dielectric fluids maintained stable optical

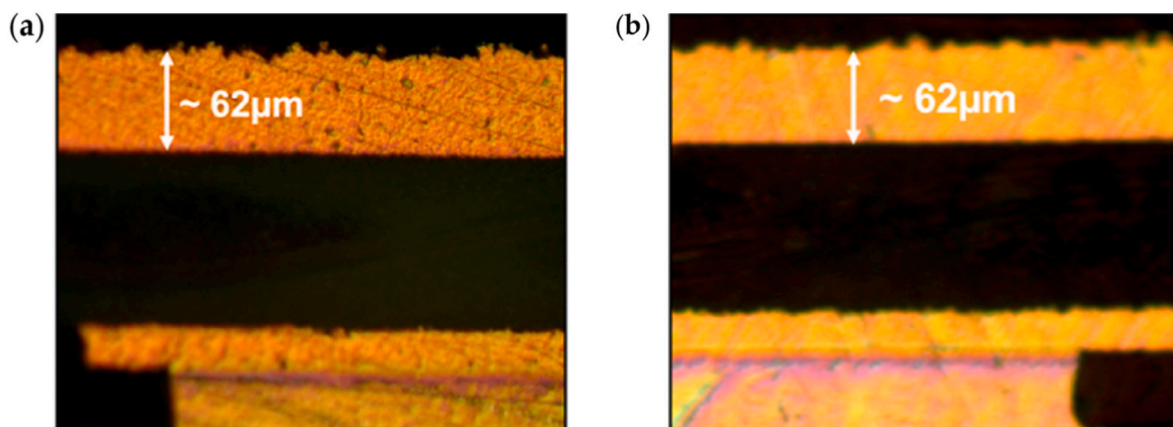


Fig. 14. Cross-sectional comparison of ball grid array (BGA) substrate layers from (a) an air-cooled server and (b) an oil-immersed server. Both configurations exhibit consistent substrate thickness ($\sim 62\ \mu\text{m}$) without evidence of delamination or structural degradation, confirming that immersion cooling does not alter substrate integrity when compatible materials are used. Figure adapted from Shah et al., [20].

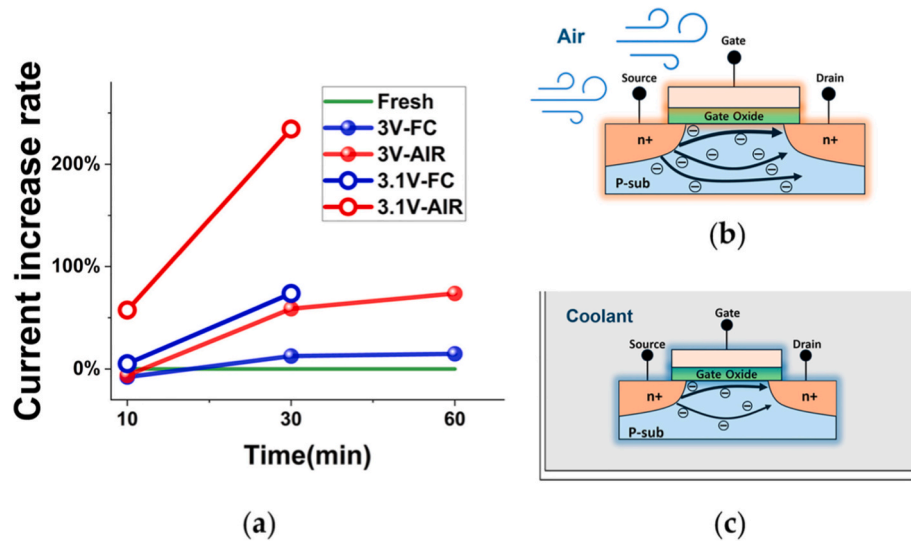


Fig. 15. Leakage current behavior under electrical stress: (a) Current increase rate at 3 V and 3.1 V for devices in air vs FC immersion; (b) schematic showing leakage pathways under air cooling; (c) schematic illustrating reduced leakage in immersion. Figure adapted from Min et al., [100].

power and insertion loss across extended test durations when optical interfaces were adequately sealed to prevent fluid ingress. No critical packaging failures were reported, validating the feasibility of SPIC for high-speed interconnects under appropriate sealing conditions.

Collectively, these findings confirm that SPIC can deliver reliability performance comparable to or exceeding that of air cooling when deployed with rigorous material qualification, comprehensive electrical stress testing, and targeted design safeguards. Long-term system qualification should integrate: (i) material compatibility testing for laminates and encapsulants, (ii) extended electrical bias and thermal aging evaluations to monitor leakage and dielectric stability, and (iii) optical performance validation for photonics modules under immersion conditions. As illustrated in Fig. 16, the interplay between operational stressors, fluid degradation mechanisms (e.g., oxidation, hydrolysis, gas evolution), and diagnostic indicators provides a structured framework

for assessing ageing in SPIC dielectric fluids.

5.2. Maintenance practices

(A) Fluid Quality and Ageing (Chemistry, Stability, Diagnostics)

Ensuring long-term reliability in single-phase immersion cooling (SPIC) systems requires proactive maintenance practices focused on fluid quality monitoring, ageing diagnostics, and early fault detection. Dielectric fluids, commonly used as immersion coolants, are subject to physicochemical degradation under prolonged thermal stress, material interaction, and contaminant ingress. These degradation mechanisms manifest as variations in moisture content, acidity, breakdown voltage (BDV), dielectric loss factor, and dissolved gas concentrations—all of which critically influence electrical insulation integrity and thermal

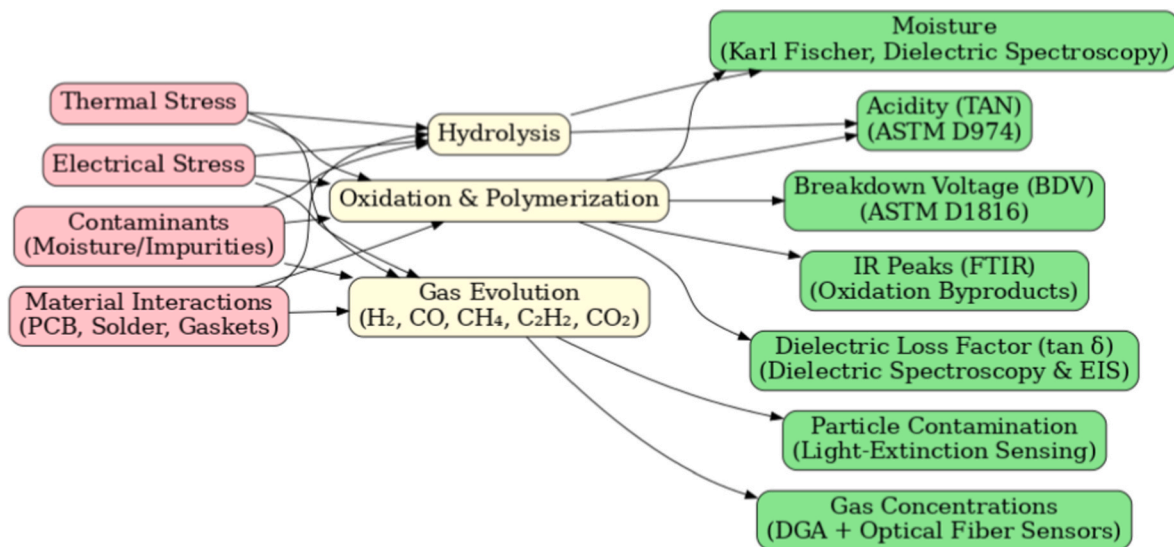


Fig. 16. Diagnostic Framework for Fluid Ageing in SPIC Systems. This figure illustrates the relationship between operational stressors (thermal/electrical loads, contaminant ingress, material interactions), fluid degradation mechanisms (oxidation, hydrolysis, gas evolution), and corresponding diagnostic parameters. Key ageing indicators include moisture (Karl Fischer titration, dielectric spectroscopy), acidity (TAN via ASTM D974), breakdown voltage (ASTM D1816), dielectric loss (tan δ), oxidation products (FTIR spectroscopy), gas evolution (DGA, optical fiber sensors), and particulate contamination (light-extinction sensing). The framework supports condition-based maintenance and early detection of insulation degradation in SPIC dielectric fluids. Diagnostic methods referenced from: Qian et al. [102]; Linde et al. [103]; Lazanas et al. [104]; Bustamante et al. [105]; Dai et al. [106]; Sun et al. [107]; Krogsøe et al. [108]; Soni et al. [109].

performance.

5.2.1. Chemical ageing diagnostics

Recent studies have demonstrated that chemical ageing markers provide a quantifiable basis for assessing coolant health. Infrared (IR) spectroscopy enables the identification of characteristic absorption peaks linked to oxidation byproducts, which correlate with dielectric loss factor and interfacial tension variations, offering early indicators of performance deterioration [102]. Complementary techniques such as dielectric spectroscopy exhibit high sensitivity to ageing-related changes, including moisture ingress and molecular mobility alterations, allowing for non-destructive and in-situ condition monitoring [103]. Additionally, electrochemical impedance spectroscopy (EIS) has emerged as a robust diagnostic method, leveraging resistive and capacitive component analysis to evaluate dielectric ageing in real time, with established interpretation frameworks supporting quantitative tracking of insulation integrity [104].

5.2.2. Gas-phase monitoring via dissolved gas analysis (DGA)

Beyond chemical stability, gaseous byproducts resulting from fluid oxidation and component outgassing can be effectively monitored through dissolved gas analysis. Conventional DGA systems, widely employed in transformer oil diagnostics (where the oils function as dielectric fluids), provide continuous measurement of critical gases such as H_2 , CO , CO_2 , CH_4 , and C_2H_2 , serving as early indicators of electrical or thermal faults [105]. Recent advancements have introduced compact optical sensing platforms, utilizing Raman spectroscopy, FTIR, tunable diode laser absorption spectroscopy (TDLAS), and photoacoustic spectroscopy (PAS), that improve gas selectivity and enable seamless integration into confined immersion environments [106]. These technologies are increasingly feasible for real-time gas monitoring in immersion cooling hardware. In addition, fiber-optic evanescent-wave absorption using tapered single-mode silica fibers enables direct in-situ dissolved gas measurement without phase separation, demonstrating a minimum detection limit of 0.015% and an ~ 11 min response time for ammonia detection [107].

5.2.3. Particulate contamination monitoring

Particle contamination represents another critical dimension in coolant health assessment. Optical methods, particularly light-extinction sensing originally developed for lubricating oil systems, have shown adaptability for dielectric immersion fluids, offering real-time quantification of particle size distribution and early detection of debris and degradation byproducts [108].

5.2.4. Integrated Diagnostic Framework

To ensure comprehensive fluid condition monitoring, a multi-modal diagnostic protocol should be implemented, incorporating.

1. Spectroscopic Analysis (IR and dielectric spectroscopy) for baseline chemical property tracking and oxidation detection.
2. Continuous DGA for gas composition profiling and thermal/electrical fault detection.
3. Electrochemical Impedance Spectroscopy (EIS) for quantitative evaluation of electrical and dielectric health.
4. Optical Particle Monitoring for contamination detection and wear debris analysis.

By integrating these diagnostic approaches, operators can detect early degradation trends, plan predictive maintenance, and extend the operational lifespan of immersion-cooled systems. This methodology parallels established best practices for transformer oil management but is adapted to the thermofluid-electronic context of data centers. In summary, these strategies define critical SPIC fluid health indicators, moisture content, acidity, BDV, dissolved gas concentration, and map them to complementary measurement methods, including FTIR

spectroscopy, dielectric spectroscopy/EIS, and optical or microfluidic DGA, ensuring real-time condition monitoring and high system reliability [109]. As illustrated in Fig. 17, the interplay between operational stressors, fluid degradation mechanisms (e.g., oxidation, hydrolysis, gas evolution), and corresponding diagnostic parameters underpins a comprehensive ageing assessment framework for SPIC systems.

(B) Contamination Prevention and Filtration (Particle Codes, Inline Sensing)

Particulate contamination in SPIC dielectric fluids accelerates wear, impairs thermal performance, and can catalyze chemical degradation. An effective strategy combines cleanliness targets, continuous inline sensing, and targeted filtration control.

5.2.5. Establishing cleanliness benchmarks

Cleanliness levels are typically expressed using ISO 4406 particle codes, which classify contamination by the number of particles $\geq 4 \mu m$, $\geq 6 \mu m$, and $\geq 14 \mu m$. For SPIC, these thresholds can be adapted from hydraulic and insulating oil practice, with specific limits set based on component sensitivity, fluid viscosity, and circulation rate. Maintaining cleanliness within these limits prolongs component and fluid life, reduces filter loading, and prevents unplanned shutdowns [110].

5.2.6. Inline optical particle counting

Light-extinction optical particle counters (OPCs) remain the primary choice for online particulate monitoring due to their real-time response and size discrimination capability [111]. Their accuracy depends on proper calibration and an understanding of error mechanisms. At low to moderate contamination, particle counts are reliable; at higher loads, shadowing and multiple scattering can cause underestimation [108]. Calibration with SPIC-compatible standards is essential, as dielectric fluids often differ in refractive index from mineral or hydraulic oils [110].

5.2.7. Complementary dielectric spectroscopy sensing

When fluid chemistry, opacity, or additive packages interfere with optical measurements, dielectric spectroscopy or electrochemical impedance spectroscopy (EIS) can serve as a robust alternative. By detecting changes in dielectric loss and permittivity, these methods can reveal contamination trends even when optical clarity is reduced. Differential or dual-channel designs can further isolate the effect of particulates from temperature or chemical ageing, providing a redundancy layer to OPC data [112].

5.2.8. Integrated monitoring and filtration control

The most effective contamination control strategy is not to monitor in isolation but to link detection directly to maintenance actions. OPC and dielectric spectroscopy data can feed into a condition-based maintenance (CBM) system, triggering filter change alerts when ISO code limits or dielectric property thresholds are exceeded. This linkage ensures filtration effort is proportional to actual contamination load, reducing unnecessary filter replacements and downtime [110,111]. This approach is illustrated in Fig. 17, which depicts a closed-loop monitoring and filtration architecture integrating light-extinction OPCs, dielectric spectroscopy, and CBM control logic to maintain fluid cleanliness in SPIC systems.

(C) Real-Time Condition Monitoring and Fault Diagnosis in Triboelectric Sensing Systems

Recent advances in triboelectric nanogenerator (TENG)-based sensing have enabled integrated, real-time condition monitoring frameworks capable of both self-powered data acquisition and intelligent fault detection. Rather than acting as passive transducers, modern TENG systems combine energy harvesting, sensing, and analytics in a

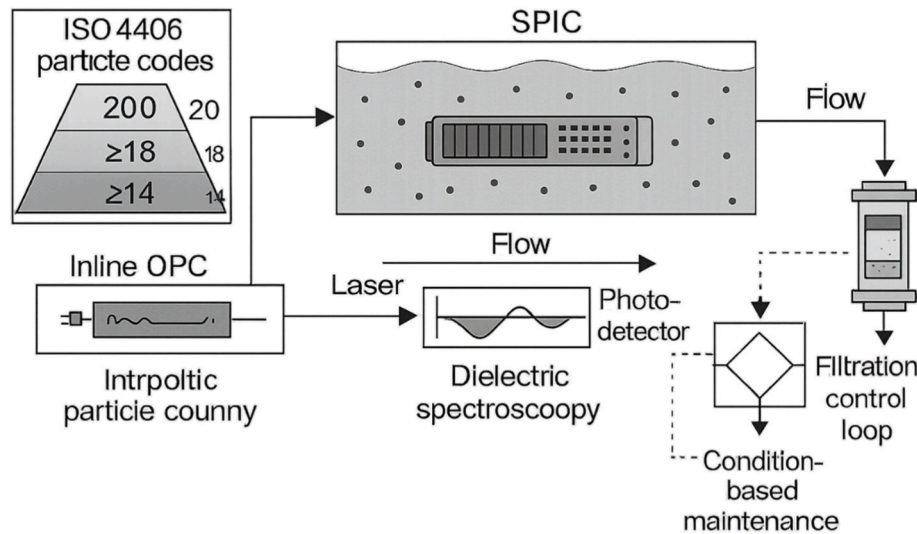


Fig. 17. This diagram presents an integrated framework for managing particulate contamination in single-phase immersion cooling (SPIC) systems. It illustrates the coordinated use of inline optical particle counters (OPCs) based on light-extinction methods for ISO 4406 cleanliness classification, and dual-channel dielectric spectroscopy to provide complementary monitoring, particularly under conditions of optical interference or complex fluid chemistry. Sensor data feeds into a condition-based maintenance (CBM) controller that actuates filtration adjustments when threshold values are exceeded, minimizing wear, fluid degradation, and unnecessary filter replacements. This closed-loop architecture reflects empirical findings from OPC error characterization, dielectric redundancy techniques, and filtration automation strategies. This figure summarizes the findings of Ng et al. [110], Krogsoe et al. [108,111], and Gong et al. [112].

single architecture, addressing key industrial needs for uninterrupted monitoring under harsh conditions and localized diagnostics at the sensing node [113].

While these systems have not yet been demonstrated in SPIC (single-phase immersion cooling) applications, their performance in lubricant and hydraulic environments suggests strong potential for adaptation. For example, rotary or sliding-mode TENGs positioned within lubricant flow paths convert mechanical motion into electrical signals whose amplitude and frequency correlate with oil properties such as viscosity, contamination level, and additive depletion. When calibrated against reference measurements (e.g., FTIR or kinematic viscosity), these triboelectric outputs can provide non-invasive oil condition estimates, avoiding issues like sensor fouling from direct contact [113,114].

To maintain signal integrity in variable environments, these TENG systems integrate multi-channel circuits for filtering noise and rotational interference. Lightweight machine learning algorithms can be deployed on edge to compare real-time signal features with historical baselines, enabling early detection of lubricant degradation or wear well before failure thresholds are reached [113].

Further sensitivity enhancements are achieved via microfluidic-assisted TENG configurations. These route a controlled lubricant microflow through a dedicated sensing channel, where shear and surface effects are optimized to increase triboelectric charge transfer. This localized measurement allows the main system to remain undisturbed while still capturing subtle changes in oil condition [113].

Beyond lubrication, the triboelectric sensing approach has been extended to structural and leakage monitoring applications. Flexible TENG arrays adhered to internal pipeline surfaces can detect anomalies in flow caused by leaks or blockages. Their distributed nature allows fault localization via phase and amplitude differences across sensor nodes, offering a scalable solution for long-span infrastructure where point sensors are impractical [115].

These innovations collectively outline a closed-loop triboelectric monitoring concept. By integrating sensing, edge analytics, and wireless data transmission, such systems align with Industry 4.0 strategies for predictive maintenance, equipment longevity, and system resilience [92,113]. Though not yet deployed in SPIC systems, they represent a promising direction for future real-time diagnostic integration.

5.3. Operational reliability (field data, failure rates, fluid degradation, long-term behavior)

Operational reliability of SPIC depends on how well systems sustain thermal performance under realistic loads, how coolants age in service, and how rack- and room-scale designs handle density increases without introducing new failure modes. Recent experimental campaigns and facility-scale studies show that SPIC can maintain tight component temperature control with reduced hot-spot occurrence across representative duty cycles, often outperforming air cooling while enabling higher allowable power density per rack [93,116]. At the heat-transfer mechanism level, comparative work on natural versus forced convection indicates that active circulation materially increases convective coefficients and suppresses temperature gradients; these gains are amplified when fin geometries and local flow paths are considered together [53]. Tank-level tests with engineered fluorinated fluids (e.g., FC-40) further demonstrate predictable thermal behavior over load sweeps, with targeted-flow strategies reducing peak junction temperatures without excessive pumping penalty [8,52].

Server- and rack-level evaluations corroborate SPIC's scalability. High-power servers immersed in dielectric fluids sustain full-load operation with headroom to throttling thresholds and measurable PUE benefits in lab and pilot settings [47]. Commercially oriented designs, such as a 48U immersion system, document how tank architecture, serviceability, and facility interfaces can be standardized for deployment at scale [49]. Physics-based models match these observations: cluster-level simulations reproduce flow distribution and hot-spot locations that align with measured data, supporting design iteration before rollout [117]. At room scale, jet-enhanced distribution and other directed-flow concepts have emerged as practical controls to mitigate stagnant zones and improve uniformity, with experimentally verified heat-transfer gains [118].

Long-term behavior also hinges on coolant stability, materials compatibility, and maintenance discipline. Review and survey work highlights that most reliability incidents originate in auxiliaries (pumps, seals, monitoring hardware) rather than the immersion heat-removal path itself, provided fluid quality and cleanliness programs are in place [6,82,83]. Design-space exploration at high energy densities showed that layout and flow-path optimization can deliver

reliability-relevant thermal improvements comparable to certain hardware changes—useful when retrofitting existing form factors [14,119]. Comparative analyses against air and hybrid baselines quantify these margins under equivalent loads and boundary conditions, informing risk-aware capacity planning [120].

Finally, operational reliability intersects directly with environmental performance. Facility-level assessments that include economizer strategies and operational scenarios find that SPIC can reduce energy consumption and associated emissions when coupled with proactive fluid maintenance and appropriate heat-rejection choices [32]. Taken together, empirical field data, lab-to-rack experiments, room-scale designs, and validated models, these works indicate that SPIC demonstrates potential for reliable, long-duration operation under well-maintained conditions when supported by targeted flow control, coolant stewardship, and deployment patterns that respect serviceability and monitoring needs [49,52,93,116].

This synthesis underscores that SPIC and TPIC both offer unique trade-offs in terms of scalability, stability, and cooling capacity, and that best practices must be tailored to facility type, workload profile, and economic constraints. These findings suggest that future improvements in SPIC system resilience may increasingly depend on integrated design-validation pipelines that consider thermal and mechanical constraints from the earliest stages. In this work, we summarize that the integration of AI-assisted verification workflows, as illustrated in Fig. 18, can contribute to operational reliability by identifying thermally induced stress points and potential failure modes early in the design process. These workflows incorporate simulation and emulation outputs into ML-based predictive analysis, which prioritizes test refinement based on component sensitivity and load behavior. In immersion contexts, this feedback mechanism may help validate how accelerator boards and

PCIe subsystems behave under sustained thermal and fluid exposure [16,49,117].

6. Challenges and limitations

Section 5 underscored how comparative evaluations clarify the trade-offs between SPIC and TPIC and distilled emerging best practices. Yet, these evaluations also reveal unresolved challenges, such as achieving thermal uniformity, ensuring coolant stability, and managing integration costs, that continue to limit widespread adoption. Section 6 therefore turns to a systematic discussion of these challenges and their implications for future deployment.

6.1. Thermal uniformity and control

One of the persistent limitations of SPIC in data centers is the difficulty in maintaining uniform temperature distribution across heterogeneous components such as CPUs, GPUs, and memory modules. As shown in Fig. 19, even under forced convection subject to an inlet temperature of 40 °C, significant thermal gradients are observed between CPUs, DIMMs, and GPUs. This highlights the significant side-effect caused by the localized hotspots, which may lead to appreciable thermal stress accordingly. This is particularly severe in high-power GPU zones, under mixed workload scenarios, especially when conventional CPUs and AI accelerators operate concurrently.

Wang et al. [13] reported that the dynamic viscosity of oil coolants critically influences convective heat transfer and flow resistance, with high-viscosity oils leading to weakened heat transfer and elevated local temperatures. Their study emphasized that although increasing inlet temperature and mass flow can improve uniformity initially, gains diminish beyond a threshold due to disproportionate increases in flow resistance. Similarly, Li et al. [122] showed that fin geometry, baffle ratio, and flowrate interactions strongly impact maximum temperature and pressure loss. Improper configurations can cause temperature differences of up to 19.8 °C across servers, highlighting the challenge of achieving thermal uniformity in practice.

At the rack scale, Kim et al. [14] demonstrated through computational analysis that immersion cooling maintained CPU and memory temperatures within safe ranges even for 800 W processors, but still exhibited variations around 9 °C between components. Compared to air cooling, immersion reduced fluctuations substantially, yet residual non-uniformity indicates that spatial heat flux mismatches remain unresolved in dense AI-CPU mixed systems. Similar findings were reported by Ge et al. [12], who noted that improper component positioning and flow maldistribution inside the tank create “thermal barriers” that degrade heat transfer for upstream devices.

As shown in Fig. 20, temperature contour plots compare T-type (left) and Z-type (right) flow configurations using FC-40 dielectric fluid. The T-type layout demonstrated improved heat dispersion and reduced hotspot formation, achieving approximately 12.6% lower thermal resistance compared to the Z-type setup, which exhibited intensified thermal gradients and localized overheating near component inlets [8]. Chen et al. [123] further quantified that dielectric liquids with lower viscosity not only improved global heat transfer but also enhanced uniformity by reducing pressure drop and eliminating stagnant regions. Likewise, Rongyang et al. [9] highlighted the sensitivity of GPU hotspot mitigation to fin parameters, with fin height and number exerting stronger influence than thickness. Suboptimal designs promoted local recirculation zones that sustained hotspot formation.

At the cabinet level, Liu et al. [45] showed that optimization of coolant and secondary water flow rates is necessary to balance energy consumption against temperature uniformity. Even with optimized designs, temperature rise differences persisted at certain load conditions, indicating the inherent trade-off between minimizing pressure drop and achieving uniform cooling.

Overall, while SPIC substantially reduces maximum component

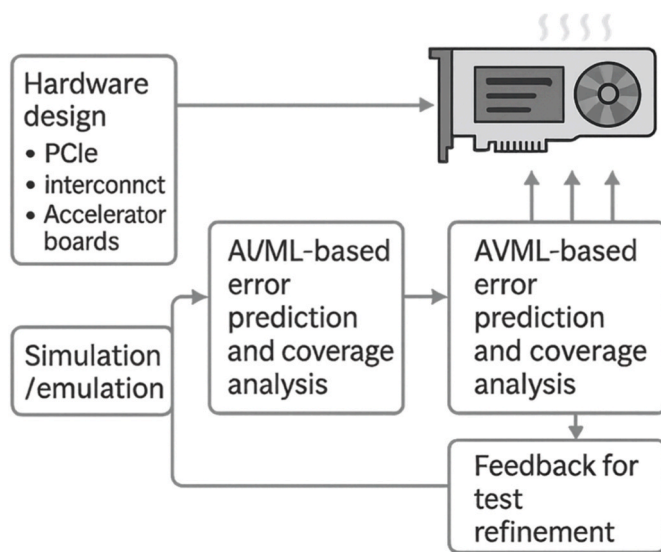


Fig. 18. AI-Assisted Verification Workflow for Hardware Design Validation. This diagram presents a flow-based architecture for accelerating hardware validation using AI/ML-driven verification strategies. The process begins with hardware design inputs—including PCIe interconnects and accelerator boards—undergoing simulation and emulation to model real-world behavior. Data generated during this phase is analyzed using machine learning algorithms for error prediction and coverage assessment. The output feeds into an AI-assisted regression pipeline that prioritizes test outcomes and identifies gaps. Thermal constraints from hardware-in-the-loop tests inform a feedback loop that refines simulation inputs iteratively. This architecture reflects system-level reliability considerations and predictive optimization techniques for high-density designs, as discussed in field and facility-level immersion cooling studies, where thermal behavior and component load under realistic conditions inform test coverage and operational decision-making (Li et al. [16]; Chang et al. [49]; David et al. [117]).

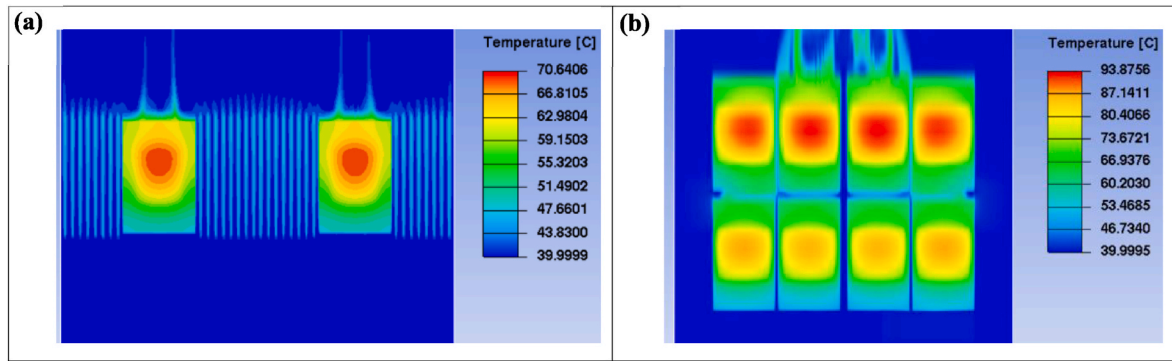


Fig. 19. Temperature Distribution in SPIC for Heterogeneous Components under Forced Convection. Temperature contours for (a) CPUs and DIMMs located on the top motherboard and (b) GPUs located on the bottom motherboard, under forced convection simulations with a 40 °C inlet fluid temperature. The figure illustrates localized hot spots in high-power components and highlights thermal gradients between component regions, underscoring the challenge of achieving uniform heat dissipation across diverse thermal loads. Figure adapted from Gupta et al. [121].

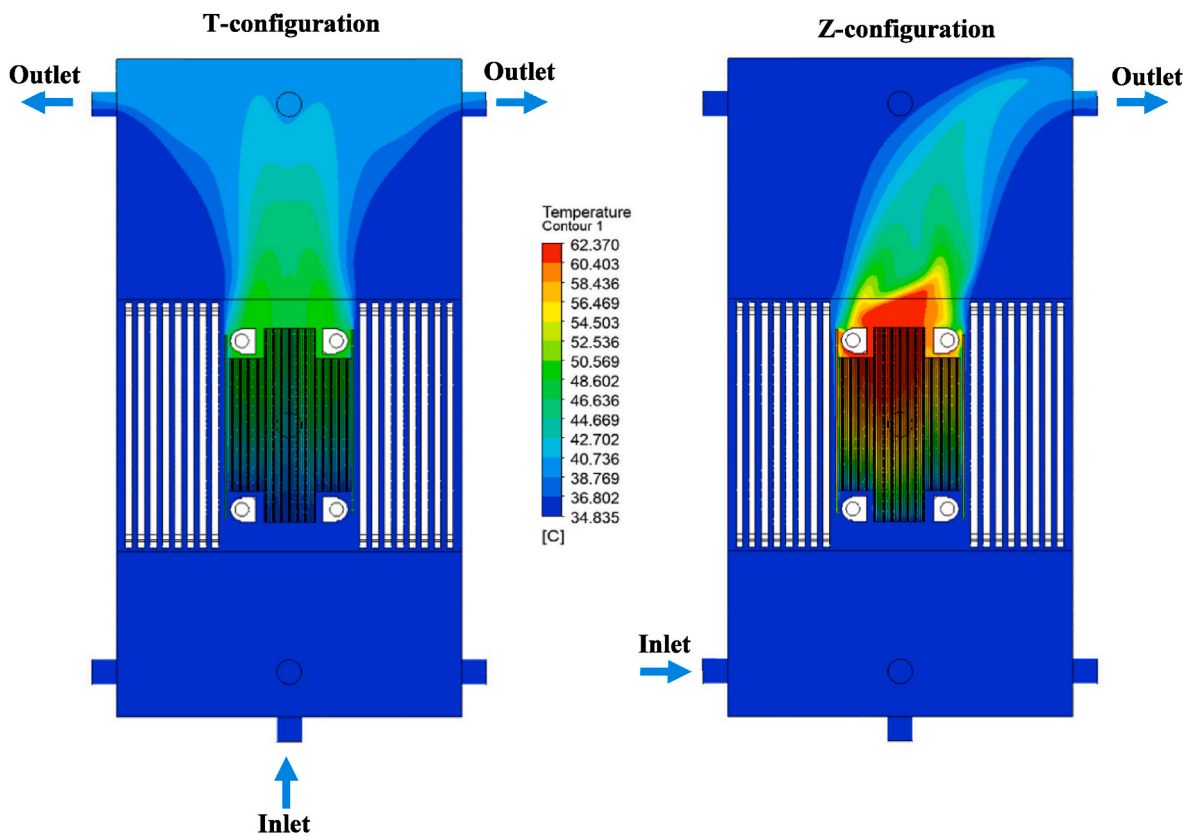


Fig. 20. Impact of Flow Configuration on Temperature Distribution in SPIC. Temperature contours for two different coolant flow strategies, T-type (left) and Z-type (right), demonstrate the effect of inlet/outlet positioning on thermal uniformity within an immersion-cooled server. The T-type configuration delivers more symmetric flow, leading to lower hotspot intensities and improved distribution across the component surface, while the Z-type exhibits concentrated hotspots due to asymmetric coolant paths. These figures, adapted from Muneeshwaran et al. [8], illustrate how flow topology significantly influences local temperature gradients and highlight the design trade-offs critical to mitigating hotspot formation in SPIC systems operating under heterogeneous workloads.

temperatures compared to air- and cold-plate cooling, ensuring thermal uniformity remains a key challenge. Variations in coolant viscosity, flow maldistribution, server architecture, and heat sink geometry lead to persistent hotspots. This issue is particularly critical for mixed AI-CPU workloads, where spatially uneven heat generation exacerbates non-uniform flow and temperature fields. Future work should prioritize multi-scale optimization, from coolant formulation to cabinet-level flow distribution, to mitigate these localized hotspots without excessive pumping penalties.

6.2. Material compatibility, fluid aging, and reliability

The long-term viability of immersion cooling systems depends not only on thermal efficiency but also on the stability of the dielectric fluids and their interactions with structural and packaging materials. Unlike traditional air- or water-cooled environments, immersion systems expose electronic assemblies and elastomeric seals to continuous contact with dielectric fluids, raising concerns about chemical compatibility, fluid aging, and reliability of the entire assembly.

Material compatibility is a key constraint in fluid selection, as even

chemically inert base oils can induce swelling, hardening, or micro-cracking in polymers and elastomers over time. Recent evaluations of polyalphaolefin (PAO) and polyol ester (POE) fluids against fluoroelastomer (FKM) indicate that PAO shows superior compatibility, with limited mass uptake and mechanical degradation compared to POE, which demonstrated more pronounced swelling and changes in tensile strength [124]. Such findings underscore the need to match fluid chemistry with sealing and gasket materials to prevent premature leakage or mechanical failure.

In parallel, the dielectric fluids themselves undergo chemical aging, driven by thermal cycling, dissolved oxygen, and electrical stress. Long-duration studies of synthetic dielectric liquids reveal gradual molecular breakdown, leading to viscosity increase, acid formation, and reduction of dielectric strength [26]. Similarly, base oils subjected to combined thermal and electrical aging show altered oxidation pathways and the formation of polar byproducts that accelerate degradation [125]. These changes compromise not only heat transfer efficiency but also increase the risk of sludge formation, deposits, and corrosive attack on metallization layers within the package.

The consequences of fluid-material interactions and chemical aging manifest directly in package-level reliability. Immersion-cooled printed circuit boards (PCBs) and electronic packages exhibit altered thermo-mechanical responses when aged in dielectric liquids, with shifts in coefficient of thermal expansion (CTE), warpage behavior, and solder joint fatigue resistance [22]. Such changes can induce early-life failures in high-density interconnects, particularly under cyclic thermal and power loading conditions typical of AI and HPC workloads.

Taken together, these findings highlight that immersion cooling reliability cannot be evaluated solely through fluid thermal properties. Instead, an integrated assessment is required that couples (i) fluid stability under realistic thermal-electrical stress, (ii) compatibility with elastomers and packaging materials, and (iii) structural reliability of PCBs and solder joints over prolonged immersion. Addressing these interdependent factors will be critical for developing standardized qualification protocols and ensuring the long-term sustainability of immersion cooling deployments.

6.3. System integration

System-level integration is one of the most significant challenges limiting the large-scale adoption of SPIC, as it requires facilities and hardware to transition from air cooling to liquid-based design. Recent analyses stress that this shift is becoming unavoidable with the growth of chip and rack powers, especially upon the surge of AI era. However, the constraints of dielectric coolants also emerge, including its lower thermophysical properties and limited flow-control mechanisms at high heat flux, and the need for coordinated optimization across system levels [83].

At the facility level, retrofit economics and operations represent a central barrier. Quantitative comparisons show immersion can cut energy consumption by about 50% and reduce occupied space by roughly two-thirds versus air cooling; moreover, multiple case experiences caution that retrofitting air-cooled data centers is often high-cost and not recommended, making purpose-built immersion facilities the more practical path [58].

Liquid-centric operation also requires additional plant systems, e.g., coolant distribution units (CDUs) and secondary/facility water loops, and attention to humidity management because immersion racks no longer provide the dehumidification effect of air-handling paths [83]. In practice, deployments place immersion among the most energy-efficient options, with reported PUE values around 1.02 to 1.04 when properly integrated [58].

At the rack/server level, integration becomes more complex under heterogeneous AI/HPC workloads. Thermal performance depends on component mix (CPUs, memory, accelerators), coolant inlet temperature, and even server height/stacking within the tank; computational

analyses show these variables materially shift temperature fields and stability [14,120].

Under extreme CPU TDP conditions approaching 800 W per processor, immersion cooling sustained safe junction temperatures ($\approx 62^\circ\text{C}$ CPUs; $\approx 44^\circ\text{C}$ memory) with only $\sim 9^\circ\text{C}$ variation. In contrast, air and hybrid cooling exhibited temperature fluctuations over 40°C and, for air, failed to hold safe operating limits [14]. Complementary evaluation at 800 W also reported that immersion reduced average CPU temperature by $\sim 22.9^\circ\text{C}$ relative to a hybrid solution [14]. These trends align with OCP/OAI guidance that recommends liquid cooling for OAM module powers beyond ~ 450 W and defines thermal test vehicles (TTVs) ≥ 700 W—with testing up to 1 kW—for evaluation (OAI System Liquid Cooling Guidelines, 2023).

Cabinet/server design further shapes device temperatures and thermal resistance. FC-40 experiments show that inlet temperature ($15\text{--}35^\circ\text{C}$), coolant flow rate ($1\text{--}3\text{ L min}^{-1}$), and suction-fan arrangements (with bypass effects) substantially affect performance; a T-type inlet/outlet reduced thermal resistance by $\sim 12.6\%$ versus a Z-type, and heat-spreader choices matter—vapor-chamber bases outperform heat pipes, both better than solid plates [8].

In summary of the foregoing discussion, the path to widespread SPIC adoption is gated by three interdependent factors: (i) retrofit costs and risks in legacy air-designed facilities [58]; (ii) readiness of facility-grade infrastructure, controls, and fluid management [83]; and (iii) rack/-server designs explicitly accommodating heterogeneous AI/HPC loads and ensuring uniform flow distribution [8,14].

7. Future trends and research directions

Building on the challenges outlined in Section 6, immersion cooling technologies are at a critical juncture. While SPIC and TPIC have demonstrated strong potential in reducing energy consumption and enabling high-density computing, their widespread adoption will depend on addressing unresolved issues of thermal uniformity, fluid reliability, and system integration.

7.1. Hybrid cooling integration

Hybrid cooling strategies are increasingly adopted as data centers shift from air-centric to liquid-centric thermal management. In these architectures, direct-to-chip cold plates, single-phase immersion cooling (SPIC), and rack/row manifolds are combined with secondary liquid loops and indirect systems to balance efficiency, operability, and scalability [29].

A widely implemented configuration is liquid-to-liquid (L2L) cooling, where cold plates or immersion baths are coupled to facility water through cooling distribution units (CDUs) and intermediate heat exchangers. Experiments have demonstrated stable operation of direct-to-chip racks at $\sim 52\text{--}53$ kW per rack, with liquid loops capturing more than 90% of the heat [126]. As shown in Fig. 21, hybrid facilities maintain thermal resilience by relying on air-side CRAH units to complement liquid loops. The figure illustrates room temperature distributions under two scenarios, the one is with normal operation with all CRAH units active and a fault condition with one CRAH unit failure, and the other one highlights how CRAHs preserve fault tolerance in hybrid cooling systems [127]. Row and rack manifolds are essential for ensuring uniform coolant distribution; validated experiments report nearly equal flow per server with small variation ($\approx 0.2\text{--}0.5$ lpm) and characterized pressure drops, enabling predictable rack-level hydraulics [128,129].

Immersion cooling further enhances energy efficiency. Recent studies report that immersion systems can reduce the cooling share of total data center energy to 3.6–13%, corresponding to PUE values between 1.037 and 1.15 [85]. However, limitations such as coolant boiling thresholds and GPU thermal throttling must be managed carefully in high-power AI workloads [85].

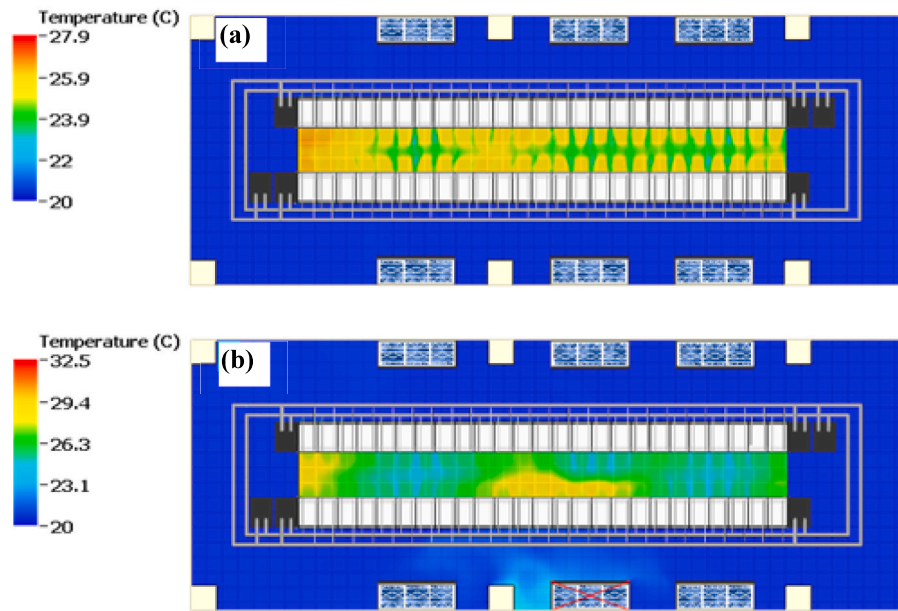


Fig. 21. Room temperature distribution at half cabinet height under hybrid cooling operation: (a) all six CRAH units operating in conjunction with liquid-to-liquid cooling, and (b) one critical CRAH unit in failure condition. The results highlight the role of air-side CRAH units in supporting liquid loops, maintaining thermal resilience, and demonstrating the fault-tolerant characteristics of hybrid cooling architectures. Figure adapted from Heydari et al. [127].

Hybridization with indirect evaporative cooling (IEC) and mechanical vapor compression (MVC) has also been investigated. A tandem IEC–MVC process demonstrated overall COP values between 4.96 and 6.05, with electricity use reduced by 19–135% compared to standalone MVC under humid and hot outdoor conditions [130]. These results highlight the resilience advantages of combining evaporative and mechanical systems where IEC performance alone is constrained. Fig. 22 provides a schematic of this combined IEC–MVC system, where outdoor air is first pre-cooled and dehumidified in the indirect evaporative cooler before entering the mechanical chiller for final conditioning. This integration underscores the resilience and efficiency benefits of coupling

evaporative and mechanical systems in climates where IEC performance alone is constrained [130].

System-level modeling further underscores hybrid cooling's scalability. Flow-network simulations of L2L distribution confirm that row and header manifolds can maintain uniform rack inlet temperatures, even under CDU redundancy scenarios such as N+1 pump failures [128]. Complementary CFD studies of European pilot data centers show that retrofitting airflow containment, rack positioning, and guided airflow control can improve overall energy efficiency by up to 75%, depending on the specific KPI used [131].

Despite these advantages, hybrid cooling introduces challenges.

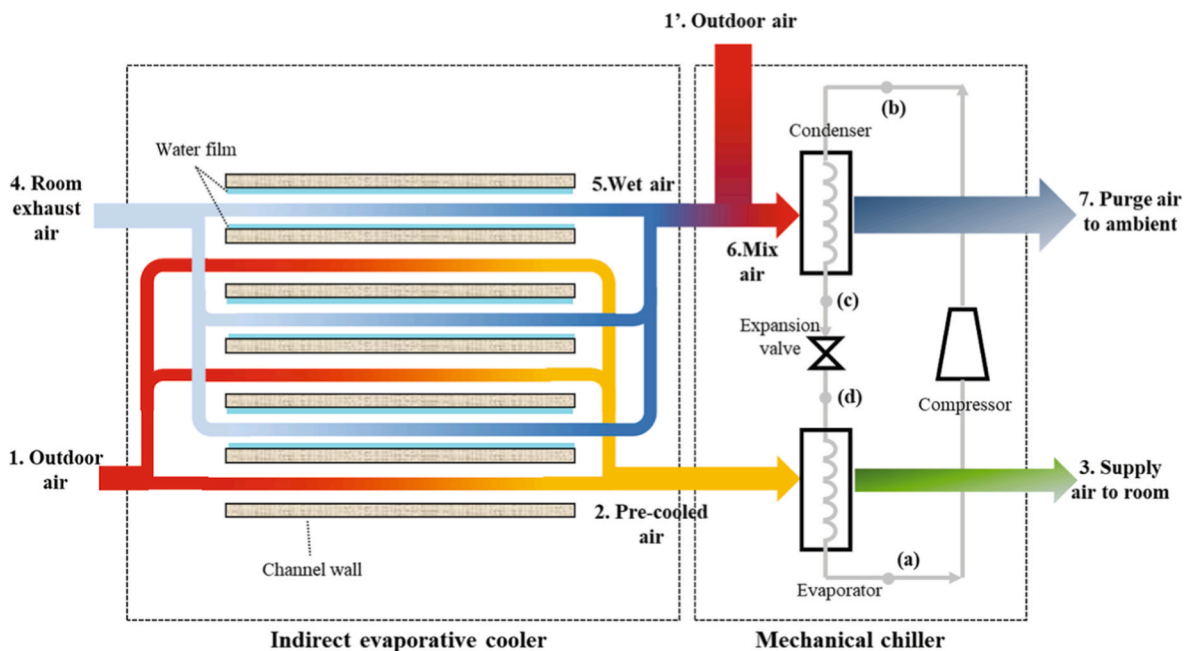


Fig. 22. Schematic of the combined indirect evaporative cooling–mechanical vapor compression (IEC–MVC) system. Outdoor air is pre-cooled and dehumidified in the IEC before entering the MVC unit for final conditioning. This tandem arrangement enhances energy efficiency and resilience in hot and humid climates where IEC performance alone is constrained. Figure adapted from Chen et al. [130].

Coolant chemistry, corrosion, and filtration remain key reliability concerns [29], and interface management between immersion tanks, liquid loops, and facility systems adds design and maintenance complexity.

Looking forward, research highlights the role of data-driven control strategies. Model-free reinforcement learning has been shown to optimize cooling setpoints, reducing energy use in large-scale data centers [132]. Similarly, artificial neural network models have been used to predict IEC-MVC performance, supporting adaptive control in dynamic conditions [130]. Together with CFD-driven retrofitting methods, these approaches point toward intelligent, hybrid infrastructures capable of sustaining the demands of high-density AI and HPC workloads while delivering significant energy efficiency gains.

7.2. Sustainable design (Net-zero, heat reuse, water-free operation)

Achieving sustainability in SPIC systems requires integration with broader net-zero strategies, including waste heat recovery, water-free operation, and circular resource use. Recent studies emphasize that immersion cooling aligns with long-term environmental goals by enabling the recovery and reuse of low-to medium-grade heat, thereby reducing dependence on fossil-based heating sources and mitigating water stress.

Jürgens et al. [133] demonstrated through regional modeling that, under favorable conditions, recovered heat from data centers could supply a significant portion of local district heating demand. Fig. 23 illustrates this integration for Frankfurt, showing how waste heat from data centers (red) and supplemental natural gas (grey) combine with electricity for heat pumps (blue) to meet flattened district heating demand through heat storage. Similarly, Pakere et al. [134] showed that dynamic integration with district networks can improve load flexibility and seasonal balancing.

Building on this, Wang et al. [135] highlighted that coupling immersion-enabled heat recovery with high-temperature heat pumps can reduce CO₂ emissions by about 85% compared with natural gas boilers, while also enhancing the economic viability of district heating integration. Fig. 24 presents the total equivalent warming impact (TEWI) of HTHPs using various low-GWP refrigerants and configurations, demonstrating both the emissions reductions and the dominant role of indirect emissions. This result underscores the dual environmental and economic benefits of integrating HTHPs into district heating networks.

Beyond direct heating applications, Corigliano et al. [136] analyzed the design and performance of Organic Rankine Cycle (ORC) systems for converting data center waste heat into electricity, underscoring their suitability for low-grade thermal sources despite modest net electric efficiencies of only a few percent. Complementing this, Yuan et al. [137] reviewed a wide range of waste-heat recovery technologies, including thermoelectric generators and ORC systems, and confirmed that electrical conversion efficiencies generally remain below 15%. They emphasized, however, that these technologies represent an important component of multi-vector energy strategies, where immersion cooling can facilitate both thermal and electrical outputs. Yuan et al. [137] further highlighted that effective operational control and integration with district-level energy networks are essential for achieving scalable deployment of such recovery solutions.

Energy reuse, however, is only one pillar of sustainable immersion design. Water intensity represents another critical dimension. Conventional air and evaporative cooling technologies rely heavily on water resources, while immersion systems eliminate this demand. Lei et al. [138] emphasized that workload type strongly influences indirect water use in data centers, while immersion cooling substantially reduces on-site water consumption (WUE-site) by avoiding evaporative cooling, thereby offering a pathway to water savings across workloads. Puentes Bejarano et al. [139] extended this analysis through environmental and social life cycle assessments, showing that the combination of waste heat reuse and water-free operation delivers not only environmental benefits but also social co-benefits such as reduced thermal pollution and improved local air quality.

Ultimately, both energy and water sustainability converge on the question of feasibility. While environmental benefits are increasingly clear, the speed of adoption depends on financial and regulatory conditions. Deymi-Dashtebayaz and Valipour-Namanlo [140] reported payback times of 3–5 years in regional feasibility studies, though outcomes vary widely with electricity pricing, heating tariffs, and capital cost structures. More recent modeling by Wang et al. [141] suggests that integrated energy system design—including participation in smart grids—can improve returns and shorten payback horizons. Still, Whitehead et al. [142] remind us that technological solutions must be matched by social acceptance and supportive regulation if net-zero outcomes are to be realized.

Taken together, these findings indicate that immersion cooling should not be viewed as an isolated thermal management solution but as

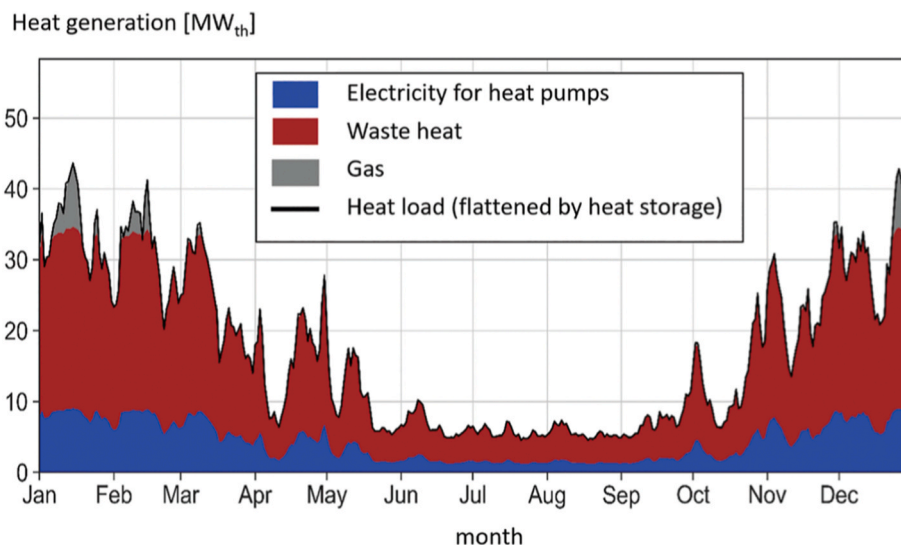


Fig. 23. Simulated annual heat generation profile for Frankfurt district heating demand under a waste-heat integration scenario. The stacked contributions represent electricity demand for heat pumps (blue), recovered waste heat from data centers (red), and supplemental natural gas (grey), with the black line indicating the flattened district heating load achieved through heat storage (Figure adapted from Jürgens et al. [133]). (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

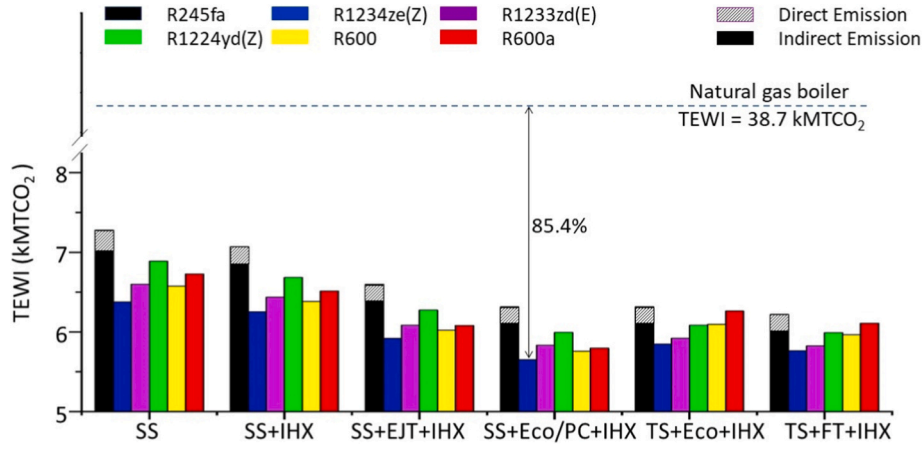


Fig. 24. Total equivalent warming impact (TEWI) of high-temperature heat pumps (HTHPs) under different configurations and low-global-warming-potential (GWP) refrigerants, compared against a natural gas boiler reference (TEWI = 38.7 kMTCO₂, 90% efficiency). Results indicate that indirect emissions dominate overall CO₂ impact, with HTHP systems achieving up to 85.4% reduction in equivalent CO₂ emissions relative to gas boilers. Figure adapted from Wang et al. [135].

part of a systemic shift toward resource efficiency. When designed to enable heat recovery, reduce water dependency, and align with economic and regulatory frameworks, SPIC can play a pivotal role in advancing carbon neutrality and circular energy use.

7.3. Intelligent control

System automation in immersion-cooled data centers increasingly relies on advanced control frameworks that can coordinate multi-level cooling operations under dynamic thermal and electrical loads. Zhao

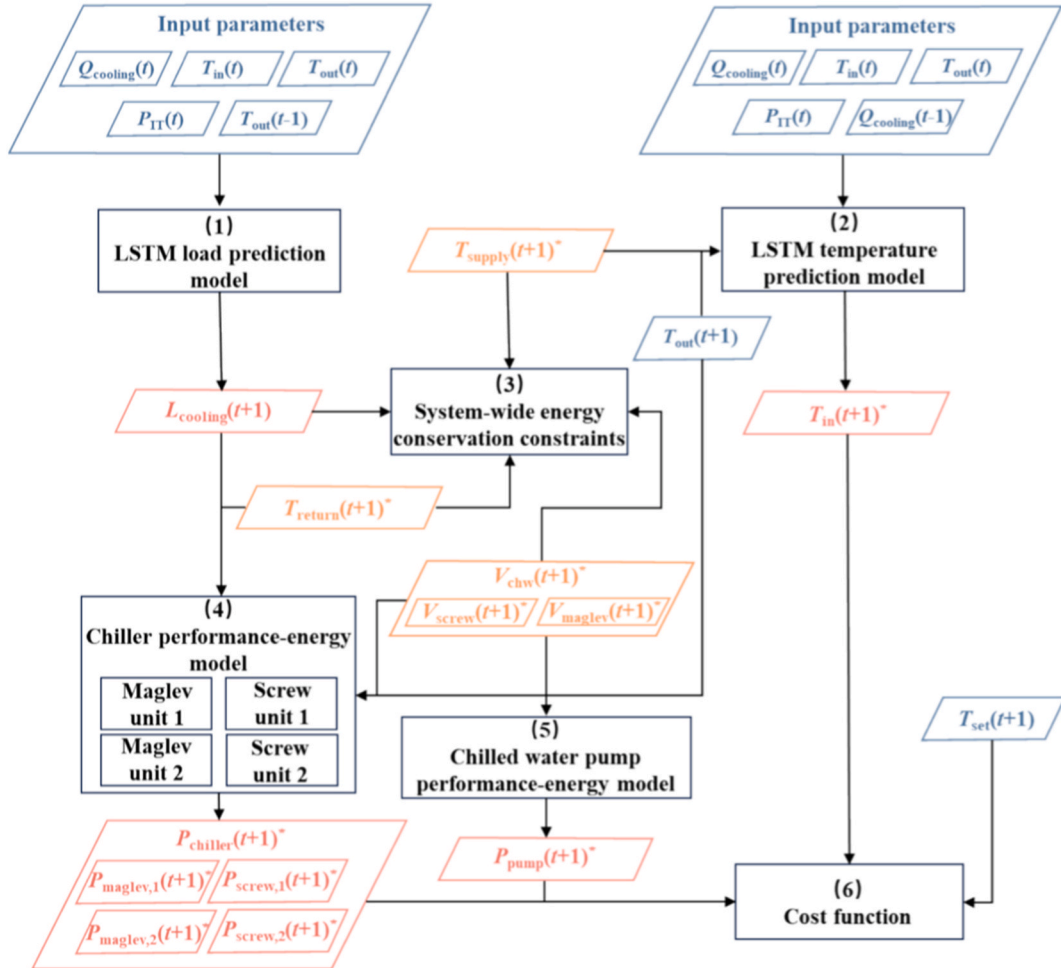


Fig. 25. Intelligent model predictive control (MPC) framework for a multi-chiller data center cooling system. The framework integrates LSTM-based load and temperature prediction with system-wide energy conservation constraints, chiller and pump performance-energy models, and a cost function to optimize both thermal stability and energy efficiency. Figure adapted from Zhao et al. [143].

et al. [143] developed a model predictive control (MPC) strategy for a multi-chiller system that integrates long short-term memory (LSTM)-based cooling load and temperature prediction models with whole-system energy conservation constraints. Their framework (Fig. 25) couples chiller and pump performance and energy models incorporated with a rolling optimization process, where a cost function balances server room temperature stability against chilled-water system energy consumption. Case study results demonstrated that this MPC approach improved server room temperature stability by 27.77% and 18.08% compared with PID and fuzzy control, respectively, while reducing chilled-water energy consumption by 11.81% and 7.58%. These findings underscore MPC's capability to deliver deterministic stability and significant energy savings in multi-chiller coordination.

Extending this line of research, Zhao et al. [144] introduced a cooling supply allocation MPC (CSA-MPC) strategy for computer room air conditioning (CRAC) systems that explicitly accounts for multi-zone thermal coupling effects. Their approach integrates a multi-zone temperature prediction model based on an attention mechanism with a particle swarm optimization algorithm to dynamically adjust supply air temperature and volume across zones. Simulation and field validation showed that the CSA-MPC strategy reduced maximum inter-zone temperature differences by 6.03 °C and decreased total cooling supply by 8.73% while maintaining uniform thermal conditions. Together, these two studies demonstrate how physics-based MPC frameworks can provide deterministic stability, fine-grained thermal control, and

measurable energy savings, establishing a strong foundation for integration with digital twins and data-driven intelligent control in data centers.

More recent efforts extend this by embedding physics-based models into digital twins and calibration frameworks, enabling predictive diagnostics, hotspot mitigation, and energy optimization across both chip and facility levels [66,145]. Zhao et al. [143] demonstrated that model predictive control (MPC) can serve as a robust foundation for intelligent cooling strategies in data centers, achieving improved temperature stability and significant energy savings compared to PID and fuzzy control. Their framework (Fig. 25) integrates LSTM-based load and temperature prediction with system-wide energy conservation constraints, chiller/pump performance models, and a cost function to balance thermal regulation and energy efficiency. This work highlights how physics-based MPC provides deterministic stability, operational transparency, and a structured optimization loop suitable for multi-chiller coordination.

A complementary strand of development centers on predictive modelling. Physics-informed neural networks (PINNs) had been applied to canonical heat transfer problems [146] and extended to large-scale borehole thermal energy storage systems [147], demonstrating their ability to generate high-fidelity thermal forecasts by embedding governing physics into machine learning frameworks. In parallel, graph neural networks (GNNs) have been shown to effectively capture airflow and heat transport in data centers, enabling accurate prediction of

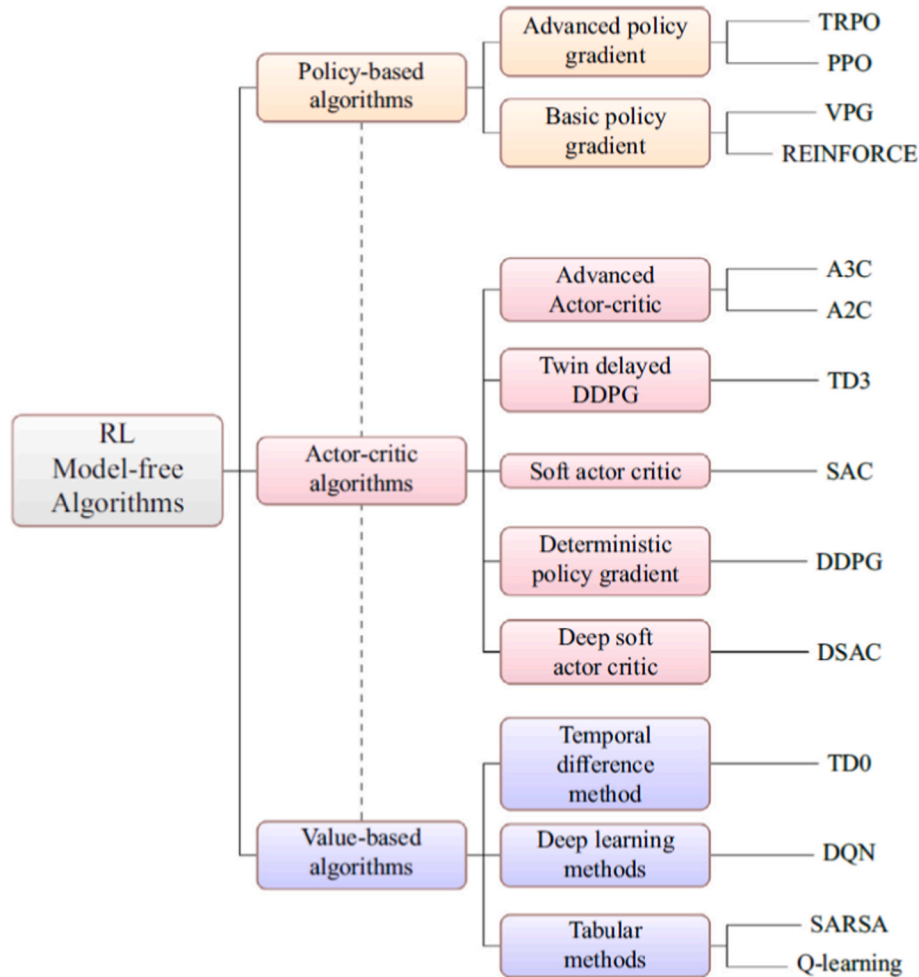


Fig. 26. RL/DRL model-free algorithms. Classification of reinforcement learning approaches that do not rely on explicit environment models. Model-free methods are divided into policy-based algorithms (e.g., REINFORCE, VPG, TRPO, PPO), actor-critic algorithms (e.g., A2C, A3C, TD3, SAC, DDPG, DSAC), and value-based algorithms (e.g., SARSA, Q-learning, DQN). These methods form the basis of deep reinforcement learning (DRL) strategies widely applied in data center cooling optimization and energy reduction. Figure adapted from Kahil et al. [65].

air-conditioning outlet temperatures across racks [148]. These data-driven methods not only enhance temperature prediction but also underpin predictive maintenance strategies. Studies on HVAC systems emphasize that integrating machine learning with IoT data enables early anomaly detection, continuous calibration of control models, and proactive intervention scheduling [149,150]. In this way, predictive modelling is evolving from a diagnostic tool into a foundation for predictive maintenance in data center cooling.

At the optimization frontier, reinforcement learning (RL) has delivered measurable cooling-energy reductions while meeting thermal constraints. Model-free and deep RL approaches demonstrate savings in operational data centers and are formulated to minimize power subject to operational temperature constraints [62,65,132]. As summarized in Fig. 26, model-free RL algorithms encompass three major families, namely the policy-based, value-based, and actor-critic methods which originate from classical tabular schemes such as SARSA and Q-learning to deep learning-driven algorithms including DQN, PPO, SAC, and DDGP. These approaches form the methodological basis of recent data center studies where RL agents learn optimal cooling strategies directly from interaction with system states, enabling both power minimization and compliance with thermal constraints. Integration with digital twins further accelerates this process: a virtual testbed allows safe, pre-deployment evaluation of cooling-plant strategies [151], while digital-twin-driven controls have shown substantial energy improvements in integrated systems [152].

Looking ahead, intelligent control for immersion cooling is likely to adopt hybrid hierarchies that combine physics-based MPC for deterministic stability, machine-learning models for real-time prediction, and RL for adaptive optimization [144]. However, deployment challenges remain: retrofits often involve heterogeneous hardware, operators may hesitate to rely on opaque models, and effective rollout requires standardized protocols for integration across IT and facility systems [153, 154].

7.4. Standardization and metrics in data center sustainability

In parallel with technological development, the deployment of immersion cooling in data centers has been accompanied by the emergence of domestic and international guidelines, industry-led specifications, and association standards aimed at ensuring safe, reliable, and interoperable implementation. These frameworks typically address electric safety under immersion conditions, material and fluid compatibility, fire and environmental considerations, maintenance practices, and facility-level integration. Notably, industry consortia and hyperscale operators have played a leading role in defining de facto requirements and best practises for immersion-cooled IT equipment, reflecting the rapid pace of adoption ahead of fully harmonized regulatory standards. While many of these efforts remain application- or region-specific, ongoing convergence across safety, reliability, and sustainability considerations highlights the growing role of standardization in supporting scalable and widespread adoption of SPIC in data center and HPC environments.

The advancement of sustainable data centers depends on standardized metrics and certification frameworks that enable cross-comparison of technologies and lifecycle performance. Traditional reliance on Power Usage Effectiveness (PUE) is increasingly seen as inadequate because it fails to capture indirect emissions, water use, or renewable energy contributions. Complementary indicators such as Carbon Usage Effectiveness (CUE), Water Usage Effectiveness (WUE), and Renewable Energy Factor (REF) have been proposed to address these gaps [148,155, 156]. Comparative reviews emphasize that while PUE has achieved global adoption, broader adoption of composite indices is necessary to reflect energy, environmental, and resource footprints more comprehensively [51].

Life Cycle Assessment (LCA) frameworks extend this benchmarking into techno-economic and environmental domains. LCA studies demonstrate the importance of evaluating both capital and operational

phases when assessing cooling and power systems [157]. For example as depicted in Fig. 27, recent work on reservoir thermal energy storage (RTES) for data center cooling applied techno-economic and lifecycle analyses, showing that RTES can cut annual electricity use by 78%, reduce cooling costs by 83%, and avoid nearly 1500 tCO₂e annually over a 20-year lifespan [158]. These findings highlight the need for harmonized LCA protocols to ensure consistency across cooling and energy system comparisons.

A further barrier to sustainable deployment is the lack of dielectric fluid certification and compatibility testing in immersion-cooled systems. Reviews underline that no unified standards exist for measuring breakdown voltage, thermal stability, or chemical reactivity [159]. Empirical studies confirm this gap, documenting degradation of substrate cores under immersion [101], accelerated aging of dielectric fluids [26], and compatibility issues of synthetic oils with fluoroelastomers [124]. Without standardized benchmarks, operators must rely on fragmented, vendor-specific claims, which complicates long-term reliability assessment.

Water sustainability metrics are another critical dimension. Analyses of Chinese data centers quantify water footprints and scarcity risks, showing that facility-level WUE varies significantly depending on cooling strategy and regional hydrology [160]. Parallel work stresses the lack of transparent and standardized reporting of water consumption in global data centers, further supporting the case for harmonized water benchmarking [161].

Finally, waste heat recovery (WHR) and circular economy indicators are gaining prominence. Reviews of WHR technologies highlight their potential to displace district heating loads and improve lifecycle sustainability, with proposals for exergy-based performance metrics [162]. Complementary studies on integrated energy systems show that coupling data centers with district heating and renewable resources could significantly enhance decarbonization outcomes if standardized accounting methods were applied [141].

In summary, the literature consistently identifies the absence of global certification frameworks and harmonized benchmarks as structural barriers. Convergence of operational metrics (PUE, CUE, WUE, REF), LCA-based sustainability indicators, standardized fluid/material protocols, and WHR reporting emerges as a research and policy priority. Without these, large-scale adoption of immersion and hybrid cooling systems risks remaining fragmented and uncertain [163].

8. Summary

This review has synthesized recent progress in SPIC for data centers and HPC, spanning dielectric fluids, system integration, and AI-enabled control strategies. Intelligent control approaches—including model predictive control, graph-based prediction, and reinforcement learning—have demonstrated measurable improvements in energy efficiency and thermal stability. These techniques, especially when integrated with digital twins, enable predictive maintenance, proactive optimization, and safe pre-deployment testing.

At the same time, sustainability has emerged as a critical dimension. Beyond PUE, broader metrics such as CUE, WUE, REF, and LCA provide more complete evaluations of environmental impact. Studies confirm that SPIC can reduce facility water usage, enhance waste heat recovery opportunities, and achieve significant lifecycle carbon savings when coupled with hybrid energy systems.

Key challenges remain in fluid durability, material compatibility, retrofit economics, and the absence of standardized certification protocols. Addressing these issues requires harmonized benchmarks that integrate operational metrics with lifecycle assessments, alongside long-term durability studies.

Building on these findings, the future evolution of SPIC is expected to be shaped by several converging trends, including intelligent control, sustainability-driven metrics, and standardization. Embedding predictive control, standardized benchmarking, and circular economy

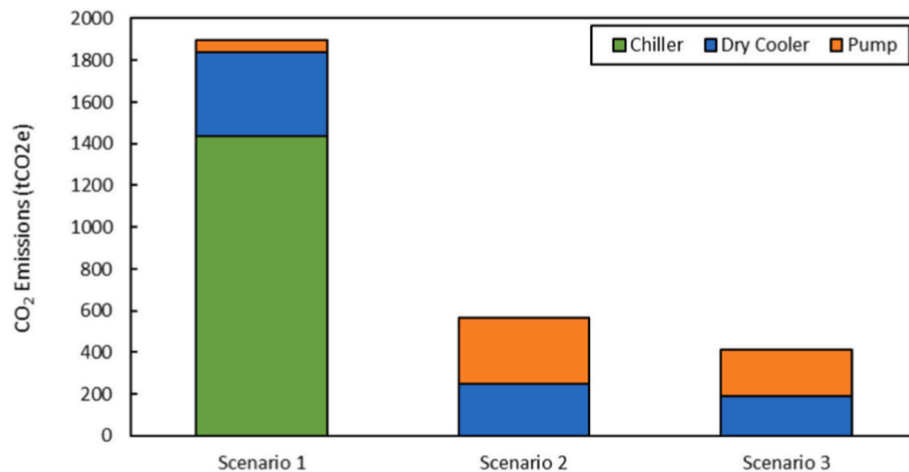


Fig. 27. Electricity use and CO₂ emissions across baseline and RTES scenarios, showing up to 78% energy savings and ~1500 tCO₂e annual reductions when reservoir thermal energy storage with heat recovery is applied. Figure adapted from Oh et al. [158].

practices will be essential to scale SPIC from pilot demonstrations to mainstream deployment in sustainable high-density computing infrastructures.

Declaration of competing interest

I am the corresponding author, Chi-Chaun Wang. There is no conflict of interest of this study.

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Data availability

Data will be made available on request.

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